

B.TECH. COMPUTER SCIENCE AND ENGINEERING -Computer Stream

Year	FIRST SEMESTER							SECOND SEMESTER						
I	Course Code	Course Name	L	T	P	C	Course Code	Course Name	L	T	P	C		
	MAT 1102	Computational Mathematics - I	3	1	0	4	MAT 1202	Computational Mathematics - II	3	1	0	4		
	PHY 1002	Applied Physics for Engineers	3	0	0	3	CHM1002	Applied Chemistry for Engineers	3	0	0	3		
	ECE 1002	Fundamentals of Electronics	2	1	0	3	ELE 1002	Fundamentals of Electrical Engineering	3	0	0	3		
	CSS 1001	Programming for Problem Solving	2	1	0	3	CIV 1002	Engineering Mechanics and Smart Buildings	2	1	0	3		
	MME 1002	Basic Mechanical Engineering Science	3	0	0	3	CSS 1002	Introduction to Object Oriented Programming	3	0	0	3		
	HUM 1001	Communication Skills in English	1	0	2	2	CIV 1003	Environmental Studies	1	0	2	2		
	HUM 1002	Universal Human Values and Professional Ethics (MLC)	1	0	0	1	CSS 1012	Data Visualization	1	0	3	2		
	HUM 1003	Human Rights and Constitution (MLC)	1	0	0	1	CSS 1013	Introduction to Object Oriented Programming Lab	0	0	3	1		
	MME 1011	Workshop Practice	0	0	3	1	MME 1012	Computer Aided Engineering Graphics	0	0	3	1		
	CSS 1011	Programming for Problem Solving Lab	0	0	3	1								
				16	3	8	22				16	2	11	22
	Total Contact Hours (L + T + P)			27			Total Contact Hours (L + T+ P)			29				

Year	THIRD SEMESTER						FOURTH SEMESTER					
II	Course Code	Course Name	L	T	P	C	Course Code	Course Name	L	T	P	C
	MAT 2101	Discrete Mathematical Structures	3	0	0	3	MAT 2201	Probability and Optimization	3	0	0	3
	CSS 2101	Data Structures	3	1	0	4	CSS 2201	Database Systems	3	1	0	4
	CSS 2102	Data Communication and Computer Networks	3	1	0	4	CSS 2202	Design & Analysis of Algorithms	3	1	0	4
	CSS 2103	Data Analytics	3	1	0	4	CSS 2203	Introduction to Artificial Intelligence	3	0	0	3
	CSS 2104	Digital Systems and Computer Organization	3	1	0	4	CSS 2204	Operating Systems	3	0	0	3
	CSS 2111	Data Structures Lab	0	0	3	1	CSS 2211	Operating Systems Lab	0	0	3	1
	CSS 2112	Digital Systems lab	0	0	3	1	CSS 2212	Database Systems Lab	0	0	3	2
							CSS 2213	Object-Oriented Software Development Lab	0	1	3	2
			15	4	6	21			15	3	9	22
Total Contact Hours (L + T + P)			25			Total Contact Hours (L + T+ P)			27			
FIFTH SEMESTER						SIXTH SEMESTER						
III	HUM 3101	Essentials of Management	3	0	0	3	HUM 3201	Engineering Economics and Financial Management	3	0	0	3
	CSS 3101	Finite Automata and Compiler Design	3	1	0	4	CSO 32xx	CS Specialization (S4)	3	0	3	4
	CSO 31xx	CS Specialization (S1)	3	0	3	4	CSO 32xx	CS Specialization (S5)	3	0	3	4
	CSO 31xx	CS Specialization (S2)	3	0	3	4	CSP 4401	Program Elective 1/ Minor 1	3	0	0	3
	CSO 31xx	CS Specialization (S3)	3	0	3	4	CSP 4402	Program Elective 2/ Minor 2	3	0	0	3
	IOE 33xx	Open Elective – 1 (MLc)	3	0	0	3	IOE 33xx	Open Elective – 2 (MLC)	3	0	0	3
			18	1	9	22			18	0	6	20
	Total Contact Hours (L + T + P)			28			Total Contact Hours (L + T + P)			24		
SEVENTH SEMESTER						EIGHTH SEMESTER						
IV	CSP 44xx	Program Elective 3/ Minor 3	3	0	0	3	CSS 4289	Internship (MLC)				1
	CSP 44xx	Program Elective 4/ Minor 4	3	0	0	3	CSS 4998	Capstone Project				12
	CSP 44xx	Program Elective 5	3	0	0	3	CSS 4999	Capstone Project (Honours)^				20
	CSP 44xx	Program Elective 6	3	0	0	3	CSS 50xx	Honours Course 1^				4
	CSP 44xx	Program Elective 7	3	0	0	3	CSS 50xx	Honours Course 2^				4
	IOE 33xx	OPEN ELECTIVE – 3 (MLC)	3	0	0	3	CSS 50xx	Honours Course 3^				4
	CSP 4199	Minor Specialization Project *				8						
		18	0	0	18/26*						13/33^	
Total Contact Hours (L + T + P)			18									

List of Specializations

A. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CSO 3101 Foundations of Computer Vision

CSO 3102 Machine Learning

CSO 3103 Parallel Computer Architecture and Programming

CSO 3201 Deep Learning and Applications

CSO 3202 Natural Language Processing

B. NETWORKS AND IOT

CSO 3111 Embedded Systems and IoT

CSO 3112 Network Protocols and Programming

CSO 3113 Cloud Computing

CSO 3211 Network Security

CSO 3212 Wireless Communication and Networking

C. DATA SCIENCE

CSO 3121 Data Privacy and Security

CSO 3122 Machine Learning

CSO 3123 Parallel Computer Architecture and Programming

CSO 3221 Deep Learning & Applications

CSO 3222 Generative AI

D. SOFTWARE ENGINEERING

CSO 3131 Software Design Technology

CSO 3132 DevOps - Using AI

CSO 3133 Full Stack Application Development

CSO 3231 Human Computer Interaction

CSO 3232 Knowledge and Data Engineering

E. CYBER SECURITY

CSO 3141 Number Theory and Cryptography

CSO 3142 Essentials of Cyber Security

CSO 3143 Incident Response and Threat Intelligence

CSO 3241 Cyber Law, Cyber Crime and Cyber Ethics

CSO 3242 Applied Cryptography

F. QUANTUM COMPUTING

CSO 3151 Fundamentals of Quantum Computing

CSO 3152 Quantum Algorithms

CSO 3153 Quantum Communications & Quantum Networks

CSO 3251 Advanced Quantum Algorithms

List of Minor Specializations

I. MULTIMODAL DATA ANALYTICS

1. CSP 4401 Information Retrieval
2. CSP 4402 Graph Analytics
3. CSP 4403 Image and Video Analysis
4. CSP 4404 Social Network Analysis

II. HEALTH CARE ANALYTICS

1. CSP 4405 Health Informatics
2. CSP 4406 Bioinformatics
3. CSP 4407 Medical Image Processing
4. CSP 4408 Biostatistics

List of Program Electives

CSP 4409 Internet of Things and Applications
CSP 4410 Augmented and Virtual Reality
CSP 4411 Conversational AI
CSP 4412 Fundamentals of iOS
CSP 4413 iOS Application Development
CSP 4414 Blockchain Technology
CSP 4415 Wireless Technologies
CSP 4416 Foundations of Quantum Computing
CSP 4417 Reinforcement Learning
CSP 4418 Mobile Application Development
CSP 4419 Foundations and Ethics of Generative AI
CSP 4420 Prompt Engineering
CSP 4421 Product Management
CSP 4422 Enterprise Data Architecture
CSP 4423 Optimization Techniques
CSP 4424 Cyber Forensics
CSP 4425 Deep Learning
CSP 4426 Machine Learning Tools & Technology
CSP 4427 Web Technologies
CSP 4428 Game Theory and Applications
CSP 4429 Explainable AI
CSP 4430 User Interface Engineering
CSP 4431 Semantic Web

III SEMESTER

MAT 2101

DISCRETE MATHEMATICAL STRUCTURES

[3 0 0 3]

Abstract syllabus:

Boolean Algebra: Partial ordering sets, Lattices—simple properties, Boolean algebras, Boolean polynomials: Disjunctive normal form (DNF) and Conjunctive normal form (CNF). Elementary configuration: Permutations and Combinations (basics), Generating function, Applications of Principle of inclusion and exclusion, Partitions, compositions, Fikes ordering of permutations. Graph theory: Basic definitions, properties, Eulerian graphs, Hamiltonian graphs, Trees and Properties, Matrices associated with graphs, Algorithms for finding shortest path. Group theory: Groups-subgroups, group of symmetries and applications, Cyclic groups, Cosets, Lagrange's Theorem. Propositional and Predicate Calculus: Well-formed formula, Connectives, Quantifications, Inference theory of Propositional and Predicate calculus. Matrix Decompositions: Prerequisites, QR, LU and SVD, and Applications.

References:

1. C. L. Liu: Elements of Discrete Mathematics, 2nd edition, 2007, Mc Graw Hill, New Delhi.
<https://pdfcoffee.com/elements-of-discrete-mathematics-liu-c-l--pdf-free.html>
2. J. P. Trembaly and R. Manohar: Discrete Mathematics Structures with application to computer science, 2012, Tata Mc Graw Hill.
https://www.academia.edu/37284735/Discrete_Mathematical_Structures_with_Applications_by_Manohar_pdf
3. E. S. Page and L. B. Wilson: An introduction to computational combinatorics, edn., 1979, Cambridge Univ. Press.
4. Narasingh Deo: Graph theory with Applications to computer science, Dover Publications, 2016. Originally published: Englewood Cliffs, New Jersey: Prentice-Hall, Inc., 1974.
<https://www.shahucollegelatur.org.in/Department/Studymaterial/sci/it/BCS/FY/book.pdf>
5. F. Harary, Graph Theory, Narosa Publishing House, New Delhi, Second edition, 1990, Reprint 2013. [https://users.metu.edu.tr/aldoks/341/Book%201%20\(Harary\).pdf](https://users.metu.edu.tr/aldoks/341/Book%201%20(Harary).pdf)
6. David C. Lay, Linear Algebra & Its Applications, Pearson Pub. 2016.
<https://home.cs.colorado.edu/~alko5368/lecturesCSCI2820/mathbook.pdf>
7. Alan Tucker, Applied Combinatorics, Wiley Publishers, 2012.
[https://www.isinj.com/mtusamo/Applied%20Combinatorics%20\(6th%20Edition\)%20by%20Alan%20Tucker%20Wiley%20\(2012\).pdf](https://www.isinj.com/mtusamo/Applied%20Combinatorics%20(6th%20Edition)%20by%20Alan%20Tucker%20Wiley%20(2012).pdf)
8. https://onlinecourses.nptel.ac.in/noc24_ma26/preview Introduction to Group Theory

CSS 2101

DATA STRUCTURES

[3 1 0 4]

Abstract syllabus:

Arrays, Sparse Matrix, Pointers and array of Pointers, Recursion, Memory allocation functions, Structures and array of structures, Linear Search, Binary Search, Merge sort and quick sort, Singly Linked List and Chains, Doubly Linked Lists, Circular Linked Lists, Linked Lists with Header Node, Applications using linked lists, Polynomials, Additional List Operations, Stacks, Queues, Circular Queues, Priority Queues and their Representation, Double Ended Queue, Input/Output Restricted Queues, Evaluation of Expression, Infix, Postfix and Prefix expressions and their conversions, Linked stacks and Linked Queues. Representation of Trees, Binary Trees, Expression tree, Binary Tree Traversals, Additional Binary Tree Operations, Threaded Binary Trees, Binary Search Trees - Definition, searching a Binary Search Tree, Inserting into and Deletion from Binary Search Tree, AVL trees, Red-Black Trees, Introduction, Definitions, Graph Representations, Depth First Search, Breadth First Search.

References:

1. Behrouz A. Forouzan, Richard F. Gilberg, A Structured Programming Approach Using C,(3e), Cengage Learning India Pvt. Ltd, India, 2007.
2. Ellis Horowitz, Sartaj Sahani, Susan Anderson and Freed, Fundamentals of Data Structures in C, (2e), Silicon Press, 2007.
3. Richard F. Gilberg, Behrouz A. Forouzan, Data structures, A Pseudocode Approach with C, (2e), Cengage Learning India Pvt. Ltd, India, 2009.
4. Tenenbaum Aaron M., Langsam Yedidyah, Augenstein Moshe J., Data structures using C, Pearson Prentice Hall of India Ltd., 2007.
5. Debasis Samanta, Classic Data Structures, (2e), PHI Learning Pvt. Ltd., India, 2010.
6. https://onlinecourses.swayam2.ac.in/cec25_hs62/preview [Introduction to Data Structures, Punjabi University, Patiala].

CSS 2102

DATA COMMUNICATION AND COMPUTER NETWORK [3 1 0 4]

Abstract syllabus:

Introduction to data communication, Digital and analog signals, Line coding, Modulation schemes, Multiplexing, Transmission media. Introduction to computer network, Protocol, Layered architecture, Topology and network devices, Data link layer services, Error detection and correction, Flow control, Access control: CSMA/CD, Ethernet Frame, Network layer services, IP addressing, IP datagram, Fragmentation, Options. Routing protocols. ARP, ICMP, Transport layer services-TCP, UDP. Application layer services- DNS, DHCP, HTTP, etc.

References:

1. Behrouz A. Forouzan, *TCP/IP Protocol Suite*, 4th Edition, Tata McGraw Hill 2017.
2. Behrouz Forouzan, *Introduction to data communication & networking (5e)*, Tata McGraw Hill, New Delhi-2017.
3. William Stallings, *Data & Computer Communications (10e)*, Pearson Education Inc., Noida, 2017.
4. Andrew S. Tanenbaum, *Computer Network*, 5th Edition Prentice Hall of India Pvt Ltd 2016.
5. Larry L. Peterson, Bruce S. Davie, *Computer Networks*, 6th Edition, 2020.
NPTEL LINK: <https://nptel.ac.in/courses/106105183>

CSS 2103

DATA ANALYTICS

[3 1 0 4]

Abstract syllabus:

Introduction to Analytics: Descriptive, Predictive, Prescriptive Analytics, steps in data analytics projects. **Data exploration:** Data sources, data collection, sampling distributions, data types, describing data using measures of central tendency, distributions, data tabulation, and visualization. case studies. **Data Preparation:** Data Cleaning, Data Imputation, Multivariate data analysis using- correlation, hypothesis tests, ANOVA, and confidence intervals. Feature Engineering, Data Integration, Data Transformations, Dimensionality reduction, PCA **Recommender Systems:** Generating Association Rules using Apriori Algorithm, Measures of Pattern Interestingness, Metrics-Support, Confidence, Lift, cosine similarity. Collaborative Filtering Techniques- User-based Similarity and Item-based Similarity. **Time Series:** basics, time zones, period arithmetic, moving window functions. Case studies

References:

1. Glenn J. Myatt, Wayne P. Johnson, *Making Sense of Data: A Practical Guide to Exploratory Data Analysis and Data Mining*, John Wiley Publication, Second Edition, 2014.
2. Jiawei Han, Micheline Kamber, Jian Pei, *Data Mining Concepts and Techniques* Morgan Kaufmann Publishers, Third Edition, 2012.
3. U. Dinesh Kumar, *Business Analytics: The Science of Data-Driven Decision Making*, Second Edition, Wiley Publications, 2021
4. Joel Grus, "Data Science from Scratch: First Principles with Python", O'Reilly Media, 2019.
5. Anil Maheshwari, *Data analytics: A comprehensive guide to data analysis and decision-making*, Wiley Publications, 2021.
6. <https://archive.nptel.ac.in/courses/110/106/110106072/> Introduction to Data Analytics.
7. https://onlinecourses.nptel.ac.in/noc21_cs45/preview Data Analytics with Python.

CSS 2104 DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [3 1 0 4]

Abstract syllabus:

Digital Systems: Simplification of logical expressions-K-Map simplification, Incompletely specified functions, Multilevel NAND and NOR Circuits. Arithmetic operations-addition of unsigned and signed numbers, BCD adder, Fast adder, Array multiplier. Combinational circuits- Multiplexer, Decoder, Encoder, Arithmetic comparison circuits. Sequential circuits- Flip-Flops, Design of synchronous sequential circuits, Ripple counter, Registers, Shift registers, Ring counter, Johnson counter, Programmable Logic Devices

Computer Organization: Instruction Set Architecture- Memory location, addresses and operations, Instructions and Instruction Sequencing, Addressing modes, RISC and CISC. Arithmetic Logic Unit- Multiplication of signed numbers, Booth's algorithm, Division, IEEE standard floating-point representation, Floating point arithmetic. Control Unit- Hardwired and micro-programmed approach for booth's multiplier. Memory System- Cache memories, Magnetic hard disk. Basic Input/Output- I/O Device Interface, Program-Controlled I/O, Interrupts. Parallel processing- Pipelining, Hardware Multithreading, Vector Processing, Shared-Memory Multiprocessors, Cache Coherence.

References:

1. Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital Logic with Verilog Design* (3e), Tata McGraw Hill 2014.
2. Morris Mano M. and Michael D. Ciletti., *Digital Design: With an introduction to the Verilog HDL* (5e), PHI Learning 2007.

3. Carl Hamacher, Zvonko Vranesic and Safwat Zaky, *Computer Organization and Embedded Systems*, (6e), McGraw Hill Publication, 2017.
4. Mohammed Rafiquzzaman and Rajan Chandra, *Modern Computer Architecture*, Galgotia Publications Pvt. Ltd., 2015.
5. Ronald J. Tocci, Neal S. Widmer and Gregory L Moss, *Digital Systems: Principles and Applications (12e)*, Pearson Education India, 2017.
6. William Stallings, *Computer Organization and Architecture*, (10e), Pearson, 2017

CSS 2111

DATA STRUCTURES LAB

[0 0 3 1]

Abstract Syllabus:

Solving Problems Using Arrays, Pointers and Dynamic Memory Allocation Functions, Ragged Arrays, Structures, Solving Problems Using Recursion (Sorting And Searching), Linked List Concepts - Singly And Doubly Linked Lists, Circular Singly And Circular Doubly Linked Lists, Linked List Applications, Implementation of Stack Using Arrays And Linked Lists, Stack Applications, Implementation of Queue Using Arrays And Linked Lists, Queue Applications, Tree Concepts - Binary Tree And Binary Search Tree, AVL Tree , and Tree Applications.

References:

1. Behrouz A. Forouzan, Richard F. Gilberg, *A Structured Programming Approach Using C*, (3e), Cengage Learning India Pvt. Ltd, India, 2007.
2. Ellis Horowitz, Sartaj Sahani, Susan Anderson and Freed, *Fundamentals of Data Structures in C*, (2e), Silicon Press, 2007.
3. Richard F. Gilberg, Behrouz A. Forouzan, *Data structures, A Pseudocode Approach with C*, (2e), Cengage Learning India Pvt. Ltd, India, 2009.
4. Tenenbaum Aaron M., Langsam Yedidyah, Augenstein Moshe J., *Data structures using C*, Pearson Prentice Hall of India Ltd., 2007.
5. Debasis Samanta, *Classic Data Structures*, (2e), PHI Learning Pvt. Ltd., India, 2010.

CSS 2112

DIGITAL SYSTEMS LAB

[0 0 3 1]

Abstract Syllabus :

Simulation of Logic Circuits Using Verilog: Simplification of Expressions using Kmap: SOP and POS Forms, Multilevel NAND, NOR Circuits, Arithmetic Circuits: Half Adder, Full Adder, Multi-Bit Adder/Subtractor, BCD Adder, Multiplexers, Decoders and Encoders, Latches and Flip-Flops: D, JK, and T Flip-Flops, Registers: Shift Register, Design of sequential circuits.

References :

1. Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital Logic with Verilog Design*, (3e), Tata McGraw Hill, 2014
2. Morris Mano M., *Digital Design*, (5e), PHI Learning 2007

IV SEMESTER

MAT 2201 PROBABILITY AND OPTIMIZATION

[3 0 0 3]

Abstract syllabus :

Axioms of probability, Baye's theorem - Applications, One dimensional and Two-dimensional random variables, mean and variance, properties, Chebyshev's inequality, Correlation Coefficient, Markov Chains, Distributions: Discrete and Continuous, Binomial, Poisson, exponential, Normal and Chi-square, Moment generating function, properties. Functions of random variables - One-Two dimensional, Jacobians. Sampling theory: Central limit theorem, Point estimation, Maximum Likelihood Estimator. Hypothesis: significance level, Chi square test. Gradients of Matrices: Useful Identities for Computing Gradients, Backpropagation and Automatic Differentiation. Constrained Optimization.

References:

1. P. L. Meyer: introduction to probability and statistical applications, 2nd edition, 1980, oxford and ibh publishing, delhi. <https://pdfcoffee.com/pl-meyerpdf-pdf-free.html>
2. Miller, freund and johnson, probability and statistics for engineers, 8th edn., phi, 2011. <http://ndl.ethernet.edu.et/bitstream/123456789/33524/1/richard%20a.%20johnson18.pdf>
3. Hogg and craig, introduction to mathematical statistics, 6th edn, 2012, pearson education, new delhi. <https://minerva.it.manchester.ac.uk/~saralees/statbook2.pdf>
4. Ross sheldon m, introduction to probability and statistics for engineers and scientists, elsevier, 2010. <https://minerva.it.manchester.ac.uk/~saralees/statbook3.pdf>

5. Marc peter deisenroth, a. Aldo faisal, cheng soon ong, mathematics for machine learning, cambridge university press, 2020. <https://mml-book.github.io/book/mml-book.pdf>
6. J. Medhi, stochastic processes, third edition, new age international, 2009. <https://pdfcoffee.com/qdownload/stochastic-processes-by-jyotiprasad-medhipdf-pdf-free.html>
7. <https://nptel.ac.in/courses/111101004> Introduction to probability theory
8. https://onlinecourses.nptel.ac.in/noc22_ma81/preview Introduction to probability theory and statistics
9. <https://nptel.ac.in/courses/111105039> Optimization.

CSS 2201

DATABASE SYSTEMS

[3 1 0 4]

Abstract syllabus:

Database-System Applications, Database Languages, Relational Databases, Database Design, Data Storage and Querying, Database Architecture, Database Schemas, Keys, File Organization , Organization of Records in Files, Relational Query Languages, Relational Operations, SQL Data Definition, SQL Data Types and Schemas, Integrity Constraints, Set Operations, Aggregate Functions, Overview of SQL Query Language, Basic Structure of SQL Queries, Join Expressions, Overview of Query Processing, Measure of Query Cost, Selection Operation, Sorting, Join Operation, Other Operations, Overview of the Design Process, The Entity-Relationship Model, Extended E-R Features, Reduction to Relational Schemas, Features of Good Relational Design, Atomic Domains and Normalization, Transaction Concept, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm.

References:

1. Silberschatz, Korth, Sudarshan, *Database System Concepts, (7e)*, McGrawHill, New York, 2011.
2. Pramod J Sadalage, Martin Fowler, *NoSQL Distilled*, Addison-Wesley, 2013.
3. Ramez Elmasri and Shamkant Navathe, Durvasula V L N Somayajulu, Shyam K Gupta, *Fundamentals of Database Systems, (6e)*, Pearson Education, United States of America, 2011.
4. Thomas Connolly, Carolyn Begg, *Database Systems – A Practical Approach to Design, Implementation and Management, (4e)*, Pearson Education, England, 2005.
5. Peter Rob, Carlos Coronel, *Database Systems–Design, Implementation and Management, (10e)*, Course Technology, Boston, 2013.
6. https://onlinecourses.nptel.ac.in/noc25_cs40/preview Introduction to Database Systems

CSS 2202

DESIGN AND ANALYSIS OF ALGORITHMS

[3 1 0 4]

Abstract syllabus:

Fundamentals of Algorithms, Important Problem Types, Analysis of algorithm efficiency. Analysis Framework: Asymptotic Notations and Basic Efficiency Classes, Mathematical Analysis of Non-recursive and Recursive Algorithms. Brute force Techniques, Divide and Conquer, Decrease and Conquer: Insertion Sort, Depth First Search, Breadth First Search, Topological Sorting. Transform and Conquer: Presorting, BST, Heapsort. Space and Time tradeoffs: Input Enhancement in String Matching. Dynamic Programming: Warshall's and Floyd's Algorithms, The Knapsack Problem. Greedy Techniques: Prim's, Kruskal's and Dijkstra's Algorithm, Huffman Trees. Coping with limitations of algorithmic power, P, NP, and NP-complete Problems, Backtracking: n-Queens problem, Hamiltonian Circuit Problem, Subset-Sum Problem. Branch and Bound: Assignment Problem, Knapsack Problem, TSP.

References:

1. Anany Levitin, *Introduction to the Design and Analysis of Algorithms*, (3e), Pearson Education, 2017
2. Ellis Horowitz and Sartaj Sahni, *Computer Algorithms/C++*, (2e), University Press, 2007.
3. Thomas H. Cormen, Charles E. Leiserson, Ronal L, Rivest, Clifford Stein, *Introduction to Algorithms*, (2e), PHI, 2006
4. Kleinberg, Jon, and Tardos, Éva. *Algorithm Design*. United Kingdom, Pearson, 2013.
5. <https://archive.nptel.ac.in/courses/106/105/106105164/>

CSS 2203**ARTIFICIAL INTELLIGENCE****[3 0 0 3]****Abstract syllabus:**

Introduction: What is Artificial Intelligence (AI)? Turing test, history and the state of the art of AI. Intelligent Agents: How agents should act, mapping, agents and environments, structure of intelligent agents. Problem Solving and Search Techniques: Search state space representation, production systems, problem and production system characteristics, uninformed and heuristic search techniques. Logical Agents: The wumpus world environment, propositional logic, propositional theorem proving, agents based on propositional logic. Using Predicate Logic: Representing instances and ISA relationship, compatible functions and predicates. Knowledge Representation: Ontological engineering, knowledge representation using predicate calculus, knowledge engineering process. Quantifying Uncertainty and Probabilistic Reasoning: Probability and Bayes' theorem, design and development of tutorials on expert systems, expert system shells, knowledge acquisition.

References:

1. Stuart Russell and Peter Norvig, Artificial Intelligence A Modern Approach (3e), Pearson Education, Third Edition, 2010.
2. Elaine Rich, Kevin Knight, Shivashankar B. Nair, Artificial Intelligence (3e), Tata McGraw Hill, 2010.
3. Saroj Kaushik, Artificial Intelligence, Cengage Learning Publications, First Edition, 2011.
4. Don W. Patterson, Introduction to Artificial Intelligence and Expert Systems, PHI Publication, 2006.
5. <https://nptel.ac.in/courses/106105079> [Title of NPTEL Course: Artificial Intelligence (Prof. P. Dasgupta), IIT Kharagpur, India]
6. <https://nptel.ac.in/courses/106102220> [Title of NPTEL Course: An Introduction to Artificial Intelligence (Prof. Mausam), IIT Delhi, India]

CSS 2204

OPERATING SYSTEMS

[3 0 0 3]

Abstract syllabus:

Introduction to operating systems, Operating system services, Operating system structure, System calls, Process management: Process concept, Threads, Inter-process communication, CPU Scheduling, Process synchronization, Handling deadlocks: Deadlock characterization, Deadlock detection, Prevention, Avoidance and recovery, Memory management: Main memory, Swapping, Contiguous memory allocation, Paging, Segmentation, Virtual memory: Demand paging, Page replacement, Thrashing, Allocating kernel memory, Storage management: File management, Disk scheduling, Case study on Unix based operating system: Design principles, Kernel modules, Basic concepts of real time operating systems: Classification of real time systems, Microkernels, Scheduling.

References:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, Operating System Concepts, (10e), Wiley, 2018.
2. William Stallings, Operating Systems: Internals and Design Principles (9e), Pearson, 2017. Phillip A Laplante, Seppo J Ovaska, Real time systems design and analysis, (4e), Wiley, 2013.
3. Rajib Mall, Real time systems: Theory and Practice, (2e), Pearson, 2009.
4. https://onlinecourses.nptel.ac.in/noc24_cs108/preview Operating System Fundamentals.

CSS 2211

OPERATING SYSTEMS LAB

[0 0 3 1]

Abstract Syllabus:

Demonstrate the working of UNIX-based operating systems by experimenting with shell commands and developing shell scripts. This includes file handling and advanced searching tools, combining the preliminary shell commands to create useful scripts. Develop applications that utilize POSIX inter-process communication system calls. Implement CPU scheduling algorithms, synchronization, memory and deadlock management tools.

References:

1. Richard Blum, Christine Bresnahan, *Linux Command Line Shell Scripting Bible*, (3e), Wiley, 2018.
2. Silberschatz A., Galvin P.B.& Gagne G., *Operating System Concepts*, (10e) Wiley, 2012.

CSS 2212**DATABASE SYSTEMS LAB****[0 0 3 1]****Abstract Syllabus:**

Data Definition Language, Data manipulation language, Basic database query operations, Integrity Constraints in SQL, Nested subqueries, Join Operations, Views, PL/SQL Basics, Exception Handling, Cursors, Stored procedures, Functions, Packages, Trigger, and project on design and development of application based on database concepts.

References:

1. Silberschatz, Korth, Sudarshan, *Database System Concepts*, (7e) McGrawHill, 2011.
2. Ivan Bayross, *SQL, PL/SQL*, (3e) , BPB Publications
3. Satish Asnani, *Oracle Database 11g*, PHI, 2010.
4. Scott Urman, *ORACLE – PL/SQL Programming*, Oracle Press.

V SEMESTER COMMON COURSES

Abstract Syllabus:

Introduction to Business, Classification of Industries, Importance of management for an engineer, Manager Roles External Environment, Social Responsibility of managers, Ethics in managing, International Business, Strategies and Policies, Strategic Planning Process & tool, Span of Management, Departmentation, Recruitment, Selection, Induction, Orientation. Leadership and Motivational tools and Techniques, Group Decision, Communication, Management Control Techniques

References:

1. Robbins, S. P., Coulter, M., & DeCenzo, D. A. (2022). Fundamentals of Management (11th ed.). Pearson.
2. Jones, G. R., & George, J. M. (2022). Essentials of Contemporary Management (9th ed.). McGraw-Hill Education.
3. Koontz, H., Weihrich, H., & Cannice, M. V. (2020). Essentials of Management: An International, Innovation, and Leadership Perspective (11th ed.). McGraw Hill Education (India) Private Limited.

Abstract syllabus:

Languages, Regular Expressions, Finite Automata – DFA, NFA. Conversion of Regular Expression to NFA, NFA to DFA. Context Free Grammars and Parsing, Derivation, Parse Trees, Push Down Automata, Introduction to Language Processors, The Structure of a Compiler, Lexical Analysis: Role of the Lexical Analyzer, Input Buffering, Specifications and Recognition of Tokens, Writing a Grammar, Top Down Parsing, Bottom Up Parsing, Introduction to LR parsing, More powerful LR Parsers, Syntax-Directed Translation: Syntax-Directed Definitions, Application of Syntax-Directed Translation, Intermediate Code Generation: Variants of Syntax Trees, Three Address Code, Code Generation: Issues in Design of Code Generator, Basic Blocks and Flow Graphs, Introduction to LEX and YACC.

References:

1. Peter Linz, *an Introduction to Formal Languages and Automata*, (6e), Jones & Bartlett Learning, 2019.
2. Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, *Compilers Principles, Techniques and Tools*, (2e), Pearson Education, 2010.

3. J E Hopcroft, Rajeev Motwani & Jeffrey D Ullman, *Introduction to Automata Theory, Languages and Computation*, (3e), Pearson Education, 2006.
4. Kenneth C. Loudon, *Compiler Construction - Principles and Practice*, (1e), Thomson, 2007.
5. John R. Levine, Tony Manson, Doug Brown, *LEX & YACC*, (2e), O Reilly Media, 2012.
6. https://onlinecourses.nptel.ac.in/noc21_cs07/preview Compiler Design
7. https://onlinecourses.nptel.ac.in/noc21_cs19/preview: Introduction to Automata, Languages and Computation

VI SEMESTER COMMON COURSE

HUM 3201 ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT [3 0 0 3]

Abstract Syllabus:

Time value of money, Interest factors for discrete compounding, Nominal & effective interest rates, Present and future worth of single, uniform, and gradient cash flow, Related problems and case studies. Bases for comparison of alternatives: present worth, future worth, annual worth and rate of return method, Replacement analysis. Concepts of breakeven analysis and its use in multiple alternative evaluation. Physical & functional depreciation, Straight line depreciation, Declining and double declining balance method of depreciation, Financial statements and its analysis using ratios, Methods of describing project risk: sensitivity analysis, break-even and scenario analysis.

References:

1. Park, C. S. (2023). *Contemporary Engineering Economics*, 7th Edition. Pearson Higher Education & Professional Group.
2. B Blank, L. T., & Tarquin, A. (2017). *Engineering Economy*. United Kingdom: McGraw-Hill Education.
3. Riggs, J. L., Bedworth, D. D., & Randhawa, S. U. (2017). *Engineering economics*. 4th Edition, Tata McGraw Hill Edition.
4. White, J. A., Case, K. E., Pratt, D. B., LaScola Needy, K., & Grasman, K. S. (2020). *Fundamentals of Engineering Economic Analysis*. United Kingdom: Wiley.

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING SPECIALIZATION

V SEMESTER

CSO 3101

FOUNDATIONS OF COMPUTER VISION

[3 0 3 4]

Abstract syllabus (theory):

Introduction to computer vision and its applications, Components of an Image Processing System, Elements of Visual Perception, Gray level transformations Filtering in spatial and frequency domain, Image transformations and Colour models, Edge Detection methods (Laplacian detectors and Canny edge detector), Points and patches, Harris corner detector, Histogram of Gradients, Difference of Gaussian detector, SIFT, Colour and Texture, Feature based alignment, least squares and RANSAC, Camera models, Camera calibration, Stereo vision, Stereo correspondence, Epipolar geometry, Optical flow, Lucas Kanade method, KLT tracking method, Mean shift method, Dense motion estimation, object detection and recognition, OCR – case study.

Abstract Syllabus (Integrated lab):

Introduction to Computer Vision Library: OpenCV Installation, Basics of Library, Image and Video Data Manipulation: Reading Images and Videos, Image Enhancement Techniques: Filtering in spatial Domain, Feature Extraction: Edge Detection, Image Descriptors such as ORB, LBP, SIFT. Finding Correspondence, Camera Calibration, Stereo Vision: Depth Estimation, Tracking: Optical Flow, KLT, Mean shift.

References:

1. Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd Edition, Springer 2022
2. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing, 4th Edition, Pearson 2018
3. David A. Forsyth and Jean Ponce, Computer Vision: A Modern Approach, PHI learning, 2015
4. Jan Erik Solem, Programming Computer Vision with Python, O'Reilly, 2012
5. Prof. M. K. Bhuyan “Computer Vision and Image Processing - Fundamentals And Applications”, NPTEL course noc25_ee13.
https://onlinecourses.nptel.ac.in/noc25_ee13/preview
6. Prof. Vineeth N Balasubramanian, “Deep Learning for Computer Vision”, NPTEL course noc21_CS93.
https://onlinecourses.nptel.ac.in/noc21_cs93/preview

CSO 3102

MACHINE LEARNING

[3 0 3 4]

Abstract syllabus (theory):

An Overview of Statistical Learning. The Trade-Off Between Prediction Accuracy and Model Interpretability, Supervised Versus Unsupervised Learning, Regression Versus Classification Problems, Assessing Model Accuracy, Overfit, Underfit. Simple Linear Regression, Gradient Descent Method, Multiple Linear Regression. An Overview of Classification, Logistic Regression, Sigmoid Function, Cost Function, Gradient Descent, Naïve Bayes, K-NN, Performance metrics: Confusion Matrix, Precision and Recall & ROC curve. Well-Posed ML problems, Data representation, The Crisp-DM Model. The Basics of Decision Trees, Measures of Impurity for Evaluating Splits in Decision Trees, ID3, C4.5, and CART Decision Trees. Maximal Margin Classifier, Support Vector Classifiers, Support Vector Machines, Support Vector Machines and Kernels and Numerical examples. Clustering Methods. Rationale, Generating Diverse Learners, Voting, Bagging, Boosting.

Abstract syllabus (Integrated lab):

Understanding basics of Machine Learning programming using python, Fundamental mathematical concepts required for Machine Learning, Preparation of data for Machine Learning algorithms and Principal Component Analysis (PCA), Naïve Bayes classifier, K-Nearest Neighbour classifier, Linear, Multilinear and Polynomial Regressions, Logistic Regression, Support Vector Machines, K-Means Clustering, Hierarchical Clustering, Decision Trees.

References:

1. Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, *An Introduction to Statistical Learning with Applications in Python*, second edition, Springer, New York, 2021
2. Gopinath Rebala · Ajay Ravi, Sanjay Churiwala, *An Introduction, to Machine Learning*, Springer 2019.
3. M. Gopal, *Applied Machine Learning*, McGraw Hill Education, 2018
4. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, second edition, Springer, 2008.
5. Parteek Bhatia, *Machine Learning with Python Principles and Practical Techniques*, Cambridge University Press, 2024.
6. <https://nptel.ac.in/courses/106106139> NPTEL: Introduction to Machine Learning, by Dr. Balaraman Ravindran IIT Madras

CSO 3103 PARALLEL COMPUTER ARCHITECTURE AND PROGRAMMING

[3 0 3 4]

Abstract syllabus (theory):

Introduction: Need of Parallelism; Forms of Parallelism (SISD, SIMD, MISD, MIMD); Moore's Law and Multicore; Amdahl's Law; Speedup and performance; Programming paradigms: Shared and Distributed; PRAM Architecture Model. Shared Memory Programming: Introduction to OpenMP; OpenMP Program constructs; Execution Model and Directives; Message Passing Programming: MPI Basics; Point-to-point communications; Collective Communications; GPU Architecture: Architecture of modern GPU; Kernel function and thread organization; CUDA program structure; Transparent Scalability and Synchronization; GPU memory organization; Memory optimizations in CUDA.

Abstract Syllabus (Integrated lab):

Implement shared memory programs using OpenMP; Implement distributed memory programs using MPI; Implement GPU programs using CUDA.

References:

1. V.Rajaraman, C. Siva Ram Murthy, "Parallel Computers Architecture and Programming" Prentice-Hall India, 2016.
2. D. Kirk and W. Hwu, "Programming Massively Parallel Processors –A Hands-on approach", Elsevier Inc.,(4e), 2023.
3. Michael J. Quinn, "Parallel Programming in C with MPI and OpenMP", McGraw Hill Edition, 2003.
4. Michael J. Quinn, "Parallel Computing" McGraw Hill Edition, 2005

NPTEL Links:

1. <https://www.open-mpi.org/doc/>
2. <https://www.openmp.org/>
3. <https://developer.nvidia.com/>

VI SEMESTER

Abstract syllabus (theory):

Introduction to Deep Learning & Architectures, Machine learning basics, Neural Networks basics, Feed Forward Neural Networks, Machine Learning Vs. Deep Learning, Representation Learning, Width Vs. Depth of Neural Networks. Activation Functions, Regularization and Optimization for Deep Learning, Convolutional Neural Networks Architectural, Popular CNN Architectures: ResNet, AlexNet. Transfer Learning: Transfer learning Techniques, Variants of CNN: DenseNet, PixelNet. Sequence Modelling: Recurrent and Recursive Nets, Recurrent Neural Networks, Bidirectional RNNs. Encoder-decoder sequence to sequence architectures, BPTT for training RNN, Long Short Term Memory Networks. Auto Encoders, Regularized Autoencoders , stochastic Encoders and Decoders, Contractive Encoders, Variational Auto Encoders, Deep Generative Models, Introduction to Transformers

Abstract Syllabus (Integrated lab):

Introduction to tensors, Computational graphs, Deep Learning Library in PyTorch, Deep Feed-forward Neural Networks, Convolutional Neural Network, Transfer Learning, Regularization for Deep Neural Networks, Recurrent Neural Networks, Long-Short Term Memory (LSTM), Encoder-Decoders, Variational Auto Encoders, Generative Adversarial Networks (GANs)

References:

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, “ Deep Learning”, MIT Press, 2017.
2. Eli Stevens, Luca Antiga, and Thomas Viehmann, Deep Learning with PyTorch, Manning, 2020
3. Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017
4. Umberto Michelucci “Applied Deep Learning. A Case-based Approach to Understanding Deep Neural Networks” Apress, 2018.
5. Kevin P. Murphy "Machine Learning: A Probabilistic Perspective", The MIT Press, 2012.
6. [Deep Learning - Course](#), By Prof. Prabir Kumar Biswas | IIT Kharagpur, 12 Weeks
7. [Deep Learning - IIT Ropar - Course](#), By Prof. Sudarshan Iyengar, Prof. Padmavati | IIT Ropar, Punjab Engineering College (Deemed to be University), 12 Weeks
8. [Machine Learning And Deep Learning - Fundamentals And Applications - Course](#), By Prof. M. K. Bhuyan | IIT Guwahati, 12 Weeks

Abstract syllabus (theory):

Introduction, Text Normalization, Edit Distance, N-gram Language Models, Part-of-Speech Tagging, Hidden Markov Models, Formal Grammars of English, Parsing with Context-Free Grammars, Statistical Parsing. Naive Bayes, Text Classification, and Sentiment, Logistic Regression, Vector Semantics and Embeddings, Neural Networks and Neural Language Models, Sequence Labeling for Parts of Speech and Named Entities, RNNs and LSTMs, Transformers and Large Language Models, Fine-tuning and Masked Language models, Prompting, In-Context Learning, and Instruct Tuning, Applications: Machine Translation and Question answering.

Abstract Syllabus (Integrated lab):

Text Normalization, Edit Distance, N-gram Language Models, Part-of-Speech (POS) Tagging, Formal Grammar and Parsing with Context-Free Grammars, Statistical Parsing, Text Classification, Vector Semantics and Embeddings, Neural Language Models, Sequence Labeling for POS and Named Entity Recognition (NER), RNNs and LSTMs, Transformers and Large Language Models

References:

1. Daniel Jurafsky & James H. Martin, *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*, (3e), Pearson, 2024.
2. Nitin Indurkha, Fred J. Damerau, *Handbook of Natural Language Processing*, Second Edition, Chapman & Hall/CRC, 2010
3. Eisenstein, Jacob, *Introduction to Natural Language Processing*, First Edition, MIT Press, 2019.
4. Uday Kamath, John Liu, and James Whitaker, *Deep Learning for NLP and Speech Recognition*, 1st Edition, Springer, 2019.
5. NPTEL: Natural Language Processing By Prof. Pawan Goyal, IIT Kharagpur
6. NPTEL: Deep Learning for Natural Language Processing By Prof. Pawan Goyal, IIT Kharagpur

NETWORKS AND IOT SPECIALIZATION

V SEMESTER

CSO 3111

EMBEDDED SYSTEMS & IOT

[3 0 3 4]

Abstract Syllabus :

An overview of Computer Architecture, overview of ARM-Cortex- M Architecture, CISC versus RISC, The RISC and ARM design philosophy, ARM addressing modes, Data transfer instructions, Arithmetic and logical instructions, Shift and rotate instructions, Branch and conditional branch instructions, Function call and return, Stack, Recursive functions, Conditional execution, Assembly language programming, Input/output I/O) programming, Timer/counter programming, I/O interfacing : LED, LCD, Keyboard, Stepper motor, ADC, PWM, UART, Nested Vectored Interrupt Controller (NVIC), External hardware interrupts, IO interrupts. Basic building blocks of an IOT Device, Exemplary Device: Raspberry Pi, interfaces, Programming Raspberry Pi with Python, IOT communications.

Abstract Syllabus (Integrated lab):

Familiarization of data transfer from code segment to data segment and from data segment to data segment, Arithmetic operations, Logical instructions, Branch instructions, Code conversion from hexadecimal to decimal and decimal to hexadecimal, Packing and unpacking of ASCII digits, Sorting using selection sort and bubble sort techniques, Searching using linear and binary search techniques, Recursion, I/O interfacing of LEDs, LCD, keyboard, 7 segment display, ADC, PWM., I2C and RTC In addition to the above list of experiments, students are required to develop a mini project using mbed LPC1768board, Programming Raspberry Pi with Python, IOT communications.

References:

1. Jonathan W.V., *Embedded systems: Real-time interfacing to ARM Cortex-M microcontrollers*, 8th Edition, Createspace Independent Publishing Platform, July 2021.
2. Wilmshurst T., *Fast and Effective Embedded System Design applying the ARM mbed*, Elsevier, 2017.
3. Jonathan W.V., *Embedded systems: Introduction to Arm(r) Cortex-M Microcontrollers*, 6th Edition, Createspace Independent publishing platform, Jan 2019.
4. UM10360, *LPC 176x/5x User Manual*, NXP Semiconductors, Rev. 3.1, 2014.
5. Joseph V., *A definitive Guide to ARM Cortex-M3 and Cortex-M4 processors*, 3rd Edition, Elsevier, 2014.
6. Muhammad A.M, Sarmad N., Sepehr N., Shujen C., *ARM Assembly Language Programming & Architecture*, 2nd Edition, Wiley, 2016.
7. NPTEL/SWAYAM Reference: https://onlinecourses.nptel.ac.in/noc22_cs93/preview

Abstract Syllabus :

Queuing model, Switching, VLAN, Inter VLAN Routing, STP, Ether channel, Stateless Address Autoconfiguration, First Hop Redundancy Protocol, WLAN, IPV6, WAN, Integrated and Differentiated Services, Multiprotocol Label Switching, Socket Introduction, Socket Address Structures, Value-Result Arguments, Elementary TCP Sockets, Network Security, Virtual Private Network and IP Security.

Abstract Syllabus (Integrated lab):

Packet Tracer- Introduction to CISCO Packet Tracer, Basic Configuration, Routing Protocols, DHCP and NAT; Socket programming-Basic, Files, Database, Multi Client- Server Environment and Application Development.

References:

1. W. Richard Stevens, Bill Fenner, Andrew M. Rudoff, *UNIX Network Programming Volume 1*, 3rd Edition, Addison Wesley 2011.
2. William Stallings, *High Speed Networks and Internet*, 2nd Edition, Pearson Education New Delhi 2002.
3. James F. Kurose and Keith W. Ross, *Computer Networking: A Top-Down Approach*, 7th Edition, 2017
4. Scott Empson and Cheryl Schmidt, *Routing and Switching Essentials*, 2014
5. Todd Lammle, *CCNA Routing and Switching Study Guide*, John Wiley & Sons, 2013
6. NPTEL/SWAYAM Reference * https://onlinecourses.nptel.ac.in/noc21_cs18/preview & NPTEL :: Computer Science and Engineering - NOC: Computer Networks and Internet Protocol

CSO 3113

CLOUD COMPUTING

[3 0 3 4]

Abstract Syllabus :

Introduction: Cloud computing delivery models and services, Challenges, Cloud Infrastructure: Amazon, Google, Microsoft Azure, Open Source Platforms, Services in Cloud: Service Oriented Architecture, REST, Cloud resource virtualization: Types of virtualization, Understanding hypervisors, Virtual Machines, Dockers Containers, Virtualization at Compute, Storage and Network level, Resource Management and Scheduling: Policies and Mechanisms, Scheduling, Business Continuity and Cloud management: Fault Tolerance, Replication Methods, Cloud Security: Virtual machine security, Access control and identity management, Cloud Tools: Eucalyptus, OpenNebula/OpenStack, CloudSim,

Abstract Syllabus (Integrated lab):

Introduction to the AWS, GoogleCloud and Microsoft Azure Portal and Services ;Setting up account and basic navigation, creation of a VM on each of the platforms and installing and running a High Level Language Program, Developing a web application and hosting on cloud. Running a database, deploying an application, creating a network on any one of these 3 platforms. Openstack, Eucalyptus installation and working. Introduction to cloud sim and installation, Running built in examples and analyzing. Running SJF Scheduling algorithm on cloud simulator.

References:

1. Dan C Marinescu, *Cloud Computing Theory and Practice*, 2nd Edition, 2017
2. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, *Mastering Cloud Computing*, 2nd Edition, McGraw Hill 2017
3. Sehgal, Naresh Kumar, and Pramod Chandra P. Bhatt, *Cloud Computing: Concepts and Practices*, 1st Edition, Springer, 2018
4. Barrie Sosinsky, *Cloud Computing Bible*, 1st Edition, Wiley Publishing Inc., 2011.
5. Anthony T. Velte, Toby J. Velte, Robert Elsenpeter, *Cloud Computing: A Practical Approach*, 1st Edition, McGraw Hill 2017
6. Mark C Chu-Carroll, *Code in the Cloud*, 1st Edition, Pragmatic Bookshelf, 2011.
7. NPTEL/SWAYAM Reference: https://onlinecourses.nptel.ac.in/noc21_cs14/preview

VI SEMESTER

CSO 3211

NETWORK SECURITY

[3 0 3 4]

Abstract Syllabus :

Introduction to Information and Network Security, Symmetric-Key Ciphers: Classical and Modern encryption techniques, Block ciphers, Advanced Encryption Standard, Uses block ciphers, Asymmetric-Key Cryptographic Ciphers, Principles of public key cryptosystems, Number theory concepts, Uses of primes, Message Integrity and Message Authentication, Cryptographic hash functions, Application of cryptographic hash functions, Digital Signature, Key Management, Entity Authentication, Transport Level Security, IP security, VPN, System Security concepts, Firewalls, Network Intrusion detection and prevention systems, SET, Multimedia Security ,Advanced Encryption Concepts.

Abstract Syllabus (Integrated lab):

Explore vulnerabilities of Windows, and Linux, Installation and maintenance of firewalls, Network-based and host-based intrusion detection systems (IDS) and prevention, Developing

a security policy for an enterprise network to enable students to acquire the experience of starting from scratch and designing a functioning security system for an enterprise. Create network scenario on various smart city models and check for vulnerability.

References :

1. William Stallings, *Cryptography and Network Security: Principles and Practice* ,7th edition,Pearson Publications, 2016.
2. Behrouz A. Forouzan, Debdeep Mukhopadhyay, *Cryptography and Network Security* ,2nd Edition (Revised), Tata McGraw-Hill Education India, 2010.
3. Borko Furht, Darko Kirovski, *Multimedia Encryption and Authentication Techniques and Applications* ,1st edition, Taylor and Francis,2019.
4. Xun Yi, Russell Paulet, and Elisa Bertino, *Homomorphic Encryption and Applications* ,1st edition, Springer Publishing Company, Incorporated, 2014.
5. Brij B. Gupta, Mamta, *Secure Searchable Encryption and Data Management*,1st edition, Taylor and Francis,2021.
6. NPTEL / SWAYAM course reference:
https://onlinecourses.nptel.ac.in/noc25_ee54/preview

CSO 3212 WIRELESS COMMUNICATION AND NETWORKING [3 0 3 4]

Abstract Syllabus :

Introduction to Wireless Systems, Electromagnetic (EM) Spectrum, Antenna and Propagation, Line of Sight and Non Line of Sight, Fading, Link Budget Analysis, Cellular Concepts, 5G Networks, Architecture of 5G Networks, 5G Spectrum, Channel assignments, Handoff strategies, Improving Coverage & Capacity, Wireless channels, Statistical channel models, Channel Coding- forward error correction (FEC) coding, Advanced Modulation and Coding techniques (MFSK, QPSK, OFDM, NOMA), Multiple access techniques, OFDM, Diversity, Beamforming, Modelling of MIMO fading channels, LPWAN - Emerging Wireless Technologies: LoRa, Sigfox, IoT, NB-IoT.

Abstract Syllabus (Integrated lab):

Basics of Network Simulation - Platform required to run network simulator; Simulating a Local Area Network ; Measuring Network Performance- Network Performance Evaluation ; Simulating a Wi-Fi Network ; Simulating a WiMAX Network; Simulating a Mobile Adhoc Network; Simulating a Wireless Sensor Network ; Setting up a Bluetooth Network ; Setting up a ZigBee Network

References:

1. Harri Holma, Takehiro Nakamura, and Antti Toskala, *5G Technology 3GPP New Radio*, John Wiley & Sons Ltd, 2020.
2. Larry Peterson and OMguz Sunay, *5G Mobile Networks: A Systems Approach*, Morgan & Claypool 2020.
3. Upena Dalal, *Wireless communication and Networks (1e)*, Oxford 2015.
4. Andrea Molisch, *Wireless Communications (3e)*, John Wiley & Sons Ltd. 2022.
5. Andrea Goldsmith, *Wireless Communications (2e)*, Cambridge University press. 2012.
6. Theodore S. Rappaport, *Wireless Communications Principles and Practice*, 2nd Edition, Pearson Education India, 2010.
7. NPTEL/SWAYAM Reference: https://onlinecourses.nptel.ac.in/noc21_ee66/preview

SOFTWARE ENGINEERING SPECIALIZATION

V SEMESTER

CSO 3131

SOFTWARE DESIGN TECHNOLOGY

[3 0 3 4]

Abstract Syllabus :

Introduction to Software Engineering, Software Processes, Agile Software Development, Requirement Engineering, Requirements Modelling (Scenario-based), Requirements Modelling (Class-based), Requirements Modelling (Behavior, Patterns, and Web/Mobile Apps), Design Concepts, Architectural Design, Pattern Based Design, Component Level Design, User Interface Design, Data Design; Software Testing Strategies, Testing Conventional Applications: W&B, Testing Web Apps and Mobile Apps, Review Techniques, Software Configuration Management, Risk Management.

Abstract Syllabus (Integrated lab):

Design and Analysis of model for small scale software development using UML tools, Verification and exploring of testing tools (CO – 2, 3, 4)

References:

1. Pressman R. S., *Software Engineering A practitioner's approach*, 9th Edition, McGraw Hill, 2020.
2. Somerville Ian, *Software engineering*, 10th Edition, Pearson Education, 2019.
3. Booch G., Rumbaugh J., Jacobson I., *The Unified Modelling Language User Guide*, 2nd Edition, Pearson, 2017.
4. Rajib Mall, *Fundamentals of Software Engineering*, 5th Edition, PHI Learning, 2018.

NPTEL Links:

Software Engineering, IIT Kharagpur: <https://nptel.ac.in/courses/106105182> Software Project Management, IIT Kharagpur: <https://nptel.ac.in/courses/106105218>

CSO 3132

DEVOPS USING AI

[3 0 3 4]

Abstract Syllabus :

Introduction to DevOps life cycle, DevOps tools, AI in DevOps, Version Control System, GIT, workflow, commands, GITHUB, branching and merging, collaborating, Continuous Integration, Jenkins, creating and scheduling job, Jenkins pipeline, Jenkinsfile, Containerizing the applications, Docker life cycle, commands, dockerhub, dockerfile, Docker volume, Docker Networking, Docker compose, Kubernetes components, Kubernetes architecture, commands, Namespace, ingress, Application deployment in kubernetes, Configuration management, Ansible architecture, Setting up master slave using ansible, Ansible playbook, modules, Vault, Notify and handler, Roles, Continuous Monitoring, AI in DevOps lifecycle, AI tools in DevOps, AI-driven Testing, AI-powered Predictive Analytics.

Abstract Syllabus (Integrated lab):

Setting up a public Git repository and exploring key Git commands such as clone, fork, pull, push, and commit for effective version control, Implementation of a continuous integration (CI) pipeline using Jenkins for web applications. Containerizing applications with Docker and deploying them on a Kubernetes cluster, Automation using Ansible, creating playbooks to configure servers for application deployment.

References:

1. Pradeep chintale, DevOps Design Patterns, First Editions, BPB Publications, 2024.
2. Scott Chacon and Ben Straub, Pro Git, 2nd Edition, Apress, 2024.
3. Mitesh Soni, *Jenkins Essentials(2e)*, Packt Publishing Limited, 2017.
4. Continuous Delivery with Docker and Jenkins, Rafal Leszko, Packt Publications, Second Edition, 2019.
5. Jamon Camisso, Hanif Jetha, Katherine Juell, *Kubernetes for Full-Stack Developers*, 2020
6. Gaurav shah, *ansible Playbook Essentials*, Packt Publishing Limited, 2015

NPTEL Links :

<https://nptel.ac.in/courses/128106012>

CSO 3133

FULL STACK APPLICATION DEVELOPMENT

[3 0 3 4]

Abstract Syllabus :

Front-End Web UI frameworks and Tools: Bootstrap 4, CSS and JavaScript component, Node.js, NPM and task runners, Progressive Web Applications, Front-End Web Development with React: JavaScript ES6, Reactstrap for Bootstrap 4-based responsive UI design, react router, Flux architecture, redux, REST API, Front-End JavaScript Framework with Angular: Typescript, Angular material, Flex-layout for responsive design, components, Data binding, Angular router, Angular animation support and Angular testing, Server-side development with NodeJS, Express and MongoDB.

Abstract Syllabus (Integrated lab):

Bootstrap 4, CSS, Task runners, Progressive Web Application with Next.js, Front-End Web Development with React, Integration of REST API, Front-End JavaScript Framework with Angular, Angular application, Data binding and components, Angular router, Angular animation, RxJS, HTTP Client in Angular with REST API, Server-side development with NodeJS, Express and MongoDB-CRUD operations and authentication.

References:

1. Shama Hoque, *Full-Stack React Projects: Modern web development using React 16, Node, Express, and MongoDB*, Ingram Publishers, 2018
2. Chris Northwood, *The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer*, 1st Edition, 2018
3. Frank Zammetti, *Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker*, 1st Edition, 2020
4. Ethan Brown, *Learning JavaScript: JavaScript Essentials for Modern Application Development*, O'Reilly Media, 2020.
5. Tal Ater, *Building Progressive Web Apps*, O'Reilly Media, 2017.
6. Jayden Reed, *Next.js by example: Step-by-step guides to building modern web apps*, Kindle ed. Independently published, 2024.

NPTEL Links:

1. Introduction to Modern Application Development, <https://nptel.ac.in/courses/106106156>
2. Modern Application Development, Persistent Computing Systems & CMI, <https://nptel.ac.in/courses/106106222>
3. Web Technology, https://onlinecourses.swayam2.ac.in/nou25_cs09/preview
4. Web Based Technologies and Multimedia Applications, https://onlinecourses.swayam2.ac.in/nou25_cs03/preview

VI SEMESTER

Abstract Syllabus:

Contexts for HCI: Human, Computer, Interaction, Design Process: Interaction Design, Design Rules, Navigation Design, Principles of good design and designers: Usability, Guidelines, Golden Rules, User Support, Patterns, Accessibility, User Experience Design: UX Design Process, Visual Design Principles and processes, UI Design and Implementation, Usability Evaluation Techniques: Different measures for evaluation, Usability heuristics and the principles of usability testing, Cognitive models: Models and Architectures; Case Study: AR/VR and Multimedia, Tools: Unity, Figma.

Abstract Syllabus (Integrated lab):

User-centered design, usability principles, interaction models. Prototyping: Low/high-fidelity prototypes, wireframing, iterative design. Research Methods: Surveys, usability testing. Accessibility: WCAG and auditing tools. Evaluation: Heuristic evaluation, cognitive walkthroughs. Advanced Topics: VR/AR interfaces, Design and evaluation of interactive systems.

References:

1. Alan Dix, Janet E. Finlay, Gregory D. Abowd, and Russell Beale, *Human-Computer Interaction*, 3rd Edition, Pearson Education India, 2017.
2. Elvis Canziba, *Hands-On UX Design for Developers*, 1st Edition, Packt, 2018.
3. Samit Bhattacharya, *Human-Computer Interaction: User-Centric Computing for Design*, 1st Edition, McGraw-Hill, 2019.
4. Ben Shneiderman, Catherine Plaisant, Maxine Cohen and Steven Jacobs, *Designing the User Interface: Strategies for Effective Human-Computer Interaction*, 6th Edition, Addison-Wesely, 2017.
5. Jeffrey Rubin and Dana Chisnell. *Handbook of Usability Testing: How to Plan, Design, and Conduct Effective Tests*. 2nd Edition, Wiley, 2014.
6. Yvonne Rogers, Helen Sharp and Jenny Preece, *Interaction Design: Beyond Human - Computer Interaction*, 5th Edition, John Wiley & Sons Inc, 2019.

NPTEL Links:

1. Introduction to Human Computer Interaction.” NPTEL, <https://archive.nptel.ac.in/courses/106/106/106106177/>
2. *Human Computer Interaction NPTEL IITm*, <https://nptel.ac.in/courses/106103115>

Abstract Syllabus :

Data Warehousing, Multi-dimensional data model, OLAP operations, Warehouse schema, Data Warehousing Architecture, Warehouse server, Metadata, OLAP Engine, Data Warehouse Backend Process. Data Preprocessing, Data cleaning, Data Integration and transformation, Data reduction, Sampling, Discretization and concept hierarchy generation, Segmentation by natural partitioning. Introduction to Data mining, Association rules mining, market based analysis, Apriori Algorithm, Dynamic item set counting algorithm, FP-tree growth Algorithm. Knowledge graphs (KG): Definition – DIKW Pyramid – Linked Data and Knowledge Graphs – Anatomy – Construction of KG: Data modelling – Integrating data – Data extraction and transformation – Embedding techniques for knowledge graphs – Building – Querying. Implementation of KG using Neo4j. Ontology engineering: Introduction, classification, development, learning. Web engineering: information distribution and integration , Semantic Web , RDF and Linked Open Data , Social Web and semantics.

Abstract Syllabus (Integrated lab):

ETL, Association rules mining Building – Querying. Implementation of KG using Neo4j, RDF, SPARQL., mini project.

References:

1. Jiawei Han and Micheline Kamber, Data Mining Concepts and Techniques (4e), Morgan Kaufmann Publishers, 2022
2. Arun K Pujari, Data Mining Techniques (4e), Universities Press India, 2016
3. Maria Keet, An Introduction to Ontology Engineering, College Publications, 2018
4. Tom Heath, Christian Bizer, Linked Data Evolving the Web Into a Global Data Space, Morgan & Claypool

NPTEL Links :

1.Data mining , prof Pabitra Mishra, IIT Kharagpur, <https://archive.nptel.ac.in>

Abstract Syllabus:

Security: Security Architecture, Security Attacks, Services and Mechanisms, Model for Data Security, Introduction to Cryptography: Classical Encryption techniques, Symmetric Encryption, Number Theory: Divisibility, Modular arithmetic, Congruences, Chinese Remainder Theorem, Public-key cryptography and Message Authentication, Hash Functions, Digital Signatures, Key Distribution and Authentication, System Security, Key Distribution and Authentication, Intrusion Prevention and Detection Systems, Cyber Security. Secure Software Development Life Cycle. **Data Privacy:** Introduction to data privacy, privacy attacks and types, access control models and types, privacy policies, their specifications, privacy in different domains-medical, financial, Anonymization and Differential Privacy: Introduction, Differential Privacy with Laplace Mechanism and Gaussian noise, Building Privacy into Data Pipelines: Design privacy measures, Data sharing workflow, Data Collection, Privacy-Aware Machine Learning and Data Science: Introduction, usage and open source libraries.

Abstract syllabus (Integrated lab):

Applied exercises in modular arithmetic, congruences, and Chinese Remainder Theorem using python libraries. Lab-based experimentation with public-key cryptography algorithms. Practical applications of message authentication, hash functions, digital signatures, and secure key distribution methods. Designing threat modelling and incident responses. Practical application of machine learning for data security using open-source python libraries

References :

1. William Stallings, Cryptography and Network Security: Principles and Practice, 7th Edition, Pearson Education, 2017.
2. William Stallings, Network Security Essentials: Applications and Standards, 6th Edition, Pearson Education, 2014.
3. Atul Kahate, Cryptography and Network Security, 3rd Edition, Tata McGraw-Hill Publishing Company Limited, 2013.
4. Katharine Jarmul, Practical Data Privacy-Enhancing Security and Privacy of Data, O'Reilly Media, Inc, April 2023.
5. Ronald Leenes, Rosamunde van Brakel, Serge Gutwirth, De Hert, Paul, Data Protection and Privacy: The Age of Intelligent Machines (Computers, Privacy and Data Protection), Hart Publishing (December 28, 2017).
6. Altice labs, Secure Software Development Life Cycle, White paper, Version 1.0, August 2023.
7. Patrick McBride and Edward P. Moser, Secure System Development Life Cycle

(SDLC), security staff, 2000.

NPTEL courses:

1. *Cryptography and Network Security*, by Dr. Debdeep Mukhopadhyay, IIT Kharagpur:
<https://nptel.ac.in/courses/106105031>
2. *NOC: Cyber Security and Privacy*, by Prof. Saji K Mathew, IIT Madras:
<https://nptel.ac.in/courses/106106248>

CSO 3122

MACHINE LEARNING

[3 0 3 4]

Abstract syllabus (theory):

An Overview of Statistical Learning. The Trade-Off Between Prediction Accuracy and Model Interpretability, Supervised Versus Unsupervised Learning, Regression Versus Classification Problems, Assessing Model Accuracy, Overfit, Underfit. Simple Linear Regression, Gradient Descent Method, Multiple Linear Regression. An Overview of Classification, Logistic Regression, Sigmoid Function, Cost Function, Gradient Descent, Naïve Bayes, K-NN, Performance metrics: Confusion Matrix, Precision and Recall & ROC curve. Well-Posed ML problems, Data representation, The Crisp-DM Model. The Basics of Decision Trees, Measures of Impurity for Evaluating Splits in Decision Trees, ID3, C4.5, and CART Decision Trees. Maximal Margin Classifier, Support Vector Classifiers, Support Vector Machines, Support Vector Machines and Kernels and Numerical examples. Clustering Methods. Rationale, Generating Diverse Learners, Voting, Bagging, Boosting.

Abstract syllabus (Integrated lab) :

Understanding basics of Machine Learning programming using python, Fundamental mathematical concepts required for Machine Learning, Preparation of data for Machine Learning algorithms and Principal Component Analysis (PCA), Naïve Bayes classifier, K-Nearest Neighbour classifier, Linear, Multilinear and Polynomial Regressions, Logistic Regression, Support Vector Machines, K-Means Clustering, Hierarchical Clustering, Decision Trees.

References:

1. Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, *An Introduction to Statistical Learning with Applications in Python*, second edition, Springer, New York, 2021
2. Gopinath Rebala · Ajay Ravi, Sanjay Churiwala, *An Introduction, to Machine Learning*, Springer 2019.
3. M. Gopal, *Applied Machine Learning*, McGraw Hill Education, 2018

4. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, second edition, Springer, 2008.
5. Parteek Bhatia, *Machine Learning with Python Principles and Practical Techniques*, Cambridge University Press, 2024.
6. <https://nptel.ac.in/courses/106106139> NPTEL: Introduction to Machine Learning, by Dr. Balaraman Ravindran IIT Madras

CSO 3123 PARALLEL COMPUTER ARCHITECTURE AND PROGRAMMING

[3 0 3 4]

Abstract syllabus (theory):

Introduction: Need of Parallelism; Forms of Parallelism (SISD, SIMD, MISD, MIMD); Moore's Law and Multicore; Amdahl's Law; Speedup and performance; Programming paradigms: Shared and Distributed; PRAM Architecture Model. Shared Memory Programming: Introduction to OpenMP; OpenMP Program constructs; Execution Model and Directives; Message Passing Programming: MPI Basics; Point-to-point communications; Collective Communications; GPU Architecture: Architecture of modern GPU; Kernel function and thread organization; CUDA program structure; Transparent Scalability and Synchronization; GPU memory organization; Memory optimizations in CUDA.

Abstract Syllabus (Integrated lab)

Implement shared memory programs using OpenMP; Implement distributed memory programs using MPI; Implement GPU programs using CUDA.

References:

1. V.Rajaraman, C. Siva Ram Murthy, "Parallel Computers Architecture and Programming" Prentice-Hall India, 2016.
2. D. Kirk and W. Hwu, "Programming Massively Parallel Processors –A Hands-on approach", Elsevier Inc.,(4e), 2023.
3. Michael J. Quinn, "Parallel Programming in C with MPI and OpenMP", McGraw Hill Edition, 2003.
4. Michael J. Quinn, "Parallel Computing" McGraw Hill Edition, 2005

References:

1. <https://www.open-mpi.org/doc/>
2. <https://www.openmp.org/>
3. <https://developer.nvidia.com/>

VI SEMESTER

CSO 3211

DEEP LEARNING AND APPLICATIONS

[3 0 3 4]

Abstract syllabus (theory):

Introduction to Deep Learning & Architectures, Machine learning basics, Neural Networks basics, Feed Forward Neural Networks, Machine Learning Vs. Deep Learning, Representation Learning, Width Vs. Depth of Neural Networks. Activation Functions, Regularization and Optimization for Deep Learning, Convolutional Neural Networks Architectural, Popular CNN Architectures: ResNet, AlexNet. Transfer Learning: Transfer learning Techniques, Variants of CNN: DenseNet, PixelNet. Sequence Modelling: Recurrent and Recursive Nets, Recurrent Neural Networks, Bidirectional RNNs. Encoder-decoder sequence to sequence architectures, BPTT for training RNN, Long Short Term Memory Networks. Auto Encoders, Regularized Autoencoders , stochastic Encoders and Decoders, Contractive Encoders, Variational Auto Encoders, Deep Generative Models, Introduction to Transformers

Abstract Syllabus (Integrated lab):

Introduction to tensors, Computational graphs, Deep Learning Library in PyTorch, Deep Feed-forward Neural Networks, Convolutional Neural Network, Transfer Learning, Regularization for Deep Neural Networks, Recurrent Neural Networks, Long-Short Term Memory (LSTM), Encoder-Decoders, Variational Auto Encoders, Generative Adversarial Networks (GANs)

References:

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, “ Deep Learning”, MIT Press, 2017.
2. Eli Stevens, Luca Antiga, and Thomas Viehmann, Deep Learning with PyTorch, Manning, 2020
3. Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017
4. Umberto Michelucci “Applied Deep Learning. A Case-based Approach to Understanding Deep Neural Networks” Apress, 2018.
5. Kevin P. Murphy "Machine Learning: A Probabilistic Perspective", The MIT Press, 2012.
6. [Deep Learning - Course](#), By Prof. Prabir Kumar Biswas | IIT Kharagpur, 12 Weeks
7. [Deep Learning - IIT Ropar - Course](#), By Prof. Sudarshan Iyengar, Prof. Padmavati | IIT Ropar, Punjab Engineering College (Deemed to be University), 12 Weeks
8. [Machine Learning And Deep Learning - Fundamentals And Applications - Course](#), By Prof. M. K. Bhuyan | IIT Guwahati, 12 Weeks

Abstract Syllabus:

Introduction to Natural Language Processing: Overview of NLP and its applications, Challenges in language understanding. **Basic Text Processing:** Tokenization, Text Normalization, Edit-Distance, Regular Expressions. **Morphology and Syntax:** English Morphology, Finite-State Morphological Parsing, Stemming, Lemmatization. **Language Models and Smoothing:** N-gram Language Models, Evaluating Language Models, Smoothing Techniques. **Vector Semantics and Word Embeddings:** Word Vectors, Word2Vec, GloVe, Semantic Similarity. **Parts of Speech and Tagging:** English Word Classes, Part-of-Speech Tagging, Hidden Markov Models. **Named Entity Recognition (NER):** NER Tagging, Sequence Labeling Models. **Transformer Architecture:** Self-Attention Mechanism, Positional Encoding, Multi-head Attention. **Pre-trained Language Models:** BERT, GPT Series, T5, Large Language Models. **Fine-Tuning and Transfer Learning:** Fine-Tuning Strategies, Parameter-Efficient Methods, Domain Adaptation. **Prompt Engineering and In-Context Learning:** Prompt Design, Zero-Shot and Few-Shot Learning, Instruction Tuning. **Applications of Large Language Models:** Chatbots and Dialogue Systems, Question Answering Systems, Summarization, Machine Translation, Text Generation. **Advanced Topics in Large Language Models:** Scaling Laws, Efficient Training and Inference, Reinforcement Learning from Human Feedback (RLHF), Model Compression (Knowledge Distillation, Quantization, Pruning), Retrieval-Augmented Generation (RAG), Multimodal Models. **Ethical Considerations and Responsible AI:** Bias and Fairness in LLMs, Privacy Concerns, Model Explainability. **Case Studies and Projects:** Developing a Custom Chatbot using GPT Models, Fine-Tuning BERT for NER, Implementing Machine Translation with Transformers.

Abstract Syllabus (Integrated lab):

Introduction to tensors, Computational graphs, Text preprocessing using Pytorch libraries, Simple NLP applications like sentiment analysis, named entity recognition, text summarization. Implement basic Transformer model in PyTorch for Neural Machine Translation, Fine-tuning Hugging Face transformer models for NLP tasks. Experiments with prompt structures, contextual cues, and few-shot learning techniques to improve pre-trained LLM model responses. Updating the knowledge base of a pretrained LLM using RAG.

References:

1. Daniel Jurafsky and James H Martin. Speech and Language Processing, 2e, Pearson Education, 2009
2. Jay Alammar and Maarten Grootendorst , *"Hands-On Large Language Models"*, O'Reilly Media, 2024

3. Lewis Tunstall , Leondro von Werra ,Thomas Wolf , *Natural Language Processing with Transformers: Building Language Applications with Hugging Face*, O'Reilly Media, 2023
4. Denis Rothman, “*Transformers for Natural Language Processing*”, Packt Publishing, 2024
5. Steven Bird, Ewan Klein and Edward Loper, *Natural Language Processing with Python*, (1e), O'Reilly Media, 2009

NPTEL Course:

1. Generative AI and Large Language Models by Naveen Kumar Bhansali, IIM Bangalore, https://onlinecourses.swayam2.ac.in/imb24_mg116/preview

CYBER SECURITY SPECIALIZATION

CSO XXXX

NUMBER THEORY AND CRYPTOGRAPHY

[3 0 3 4]

Abstract Syllabus:

INTRODUCTION: Number Theory Concepts, Divisibility and Factorization: Divisibility, Greatest Common Divisors, Euclidean Algorithm, Primes: Prime Numbers, Unique Prime Factorization. The Theory of Congruences: The concept of congruences, Congruence Classes, Applications of Congruences, solving (single) linear congruence, Solving system of linear congruence, the Chinese Remainder Theorem. Fermat's Theorem and Euler's Generalization: Fermat's Little Theorem. The general case: Euler's theorem. Algebraic structures: The Euclidean Algorithm, Finite Fields of The Form $GF(p)$. Linear Congruence. Cryptography: Introduction Terminology, Classical Cryptography, types of traditional ciphers, Introduction to security and security of traditional ciphers, Introduction to security issues of modern ciphers, Symmetric-key cryptography, Information Security - Confidentiality, Integrity & Availability – Authentication, Authorization & Non- Repudiation, Symmetric Ciphers:: Conventional Encryption: Attacks on Encryption Schemes, Modes of Operation, Multiple Encryption, Introduction to DES, Triple- DES, AES,RC4.

Abstract Syllabus (Integrated lab):

Implementation of selected number theory algorithms and traditional ciphers such as shift cipher, affine cipher, vignere cipher etc.,,Demonstration of symmetric conventional cryptographic techniques, Demonstration of symmetric classic cryptographic techniques ,Traditional Ciphers, Design and implementation of homomorphic encryption techniques, Demonstration and implementation of secure communication using standard

References :

1. William Stallings, Cryptography and Network Security: Principles and Practice, (7e), Prentice Hall, 2020.
2. Behrouz A. Forouzan and Debdeep Mukhopadhyay, Cryptography and Network Security, (2e), McGraw Hill, 2008.

CSO XXXX

ESSENTIALS OF CYBER SECURITY

[3 0 3 4]

Abstract Syllabus:

Introduction to Security Trends: The Computer Security Problem - Targets and Attacks - Approaches to Computer Security - Ethics - Basic Security Terminology - Security Models. Operational and Organizational Security: Policies, Procedures, Standards, and Guidelines - Security Awareness and Training - Interoperability Agreements - The Security Perimeter - Physical Security - Environmental Issues - Wireless -Electromagnetic Eavesdropping - People—A Security Problem - People as a Security Tool. Cryptography: introduction to modern cryptographic Algorithms – Introduction to Hashing Functions - Symmetric Encryption - Asymmetric Encryption - Quantum Cryptography- Introduction to crypto analysis. Authentication and Remote Access: User, Group, and Role Management - Password Policies - Single Sign-On - Security Controls and Permissions - Preventing Data Loss or Theft. The Remote Access Process - Remote Access Methods. Intrusion detection. Standards/frameworks for cyber security, security standards, risk management, cyber security strategic thinking and incident handling

Abstract Syllabus (Integrated lab):

Understanding Cybersecurity Tools, Packet Sniffing and Analysis, Intrusion Detection and Prevention, Implementing Hashing Algorithms, Quantum Cryptography Simulation, Password Cracking and Security Analysis, Web Application Security Testing.

References :

1. Anand Shinde, “Introduction to Cyber Security Guide to the World of Cyber Security”, Notion Press, 2021.
2. Nina Godbole, Sunit Belapure, “Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives”, Wiley Publishers, 2011.

CSO XXXX INCIDENT RESPONSE AND THREAT INTELLIGENCE [3 0 3 4]

Abstract Syllabus:

Introduction to Incident Response and Threat Landscape: Overview of IR lifecycle phases, Incident classification and severity levels, Legal and regulatory considerations. Types of threats: APTs, ransomware, insider threats, Threat intelligence sources and feeds, Case studies of notable incidents. Log Analysis and Event Correlation: Identifying anomalies and patterns. SIEM tools and log aggregation, Collecting volatile data, Memory analysis and timeline creation, Prioritizing incidents, Assessing impact and scope. Developing hypotheses, Leveraging threat intelligence, Conducting network and host-based hunts. Containment and Eradication: Isolating affected systems. Removing malware and persistence mechanisms. Attribution and TTPs: Understanding nation-state adversaries. Tactics, techniques, and procedures (TTPs). Creating Incident Reports: Communicating findings to stakeholders. Legal and compliance requirements.

Abstract Syllabus (Integrated lab):

Case studies of notable incidents. Log Analysis and Event Correlation: Identifying anomalies and patterns. SIEM tools and log aggregation, Collecting volatile data, Memory analysis and timeline creation.

References :

Incident Response & Computer Forensics (Third Edition), Jason T. Luttgens, Matthew Pepe, Kevin Mandia, McGraw-Hill Education, Third Edition (Published in 2014), ISBN-10: 0071798684

CSO XXXX CYBER LAW, CYBER CRIME AND CYBER ETHICS [3 0 3 4]

Abstract Syllabus:

Cybercrimes and related offences and penalties: Introduction to Cybercrimes, Classification of cybercrimes, Distinction between cyber-crime and conventional crimes, Provisions in Indian Laws in dealing with Cyber Crimes and its critical analysis, Information Technology Act, 2000, Penalties under IT Act, Offences under IT Act. Legal Recognition of Electronic Records and Electronic Evidence, Digital Signature Certificates, Securing Electronic records and secure digital signatures, Duties of Subscribers - Role of Certifying Authorities. Cyber- crimes under the Information Technology Act,2000 - Cyber-crimes under International Law - Hacking Child Pornography, Cyber Stalking. Copyrights, Software, Copyrights vs Patents debate, Authorship and Assignment Issues, Copyright in Internet, Multimedia and Copyright issues - Software Piracy. History, Overview of developments in Information Technology and Defining E-Commerce, Understanding Ethical, Social and Political issues in E- Commerce.

Abstract Syllabus (Integrated lab):

Legal Analysis of Digital Contracts, Tracing Cybercrime Incidents, Analyzing Digital Evidence, Simulating a Phishing Attack Scenario, Analyzing Ethical Issues in E-Commerce, Understanding Copyright and Piracy

References :

1. The Information Technology Act, 2000 Bare Act with Short Notes, Universal Law Publishing Co., New Delhi
2. Publishing Co., New Delhi
3. Justice Yatindra Singh: Cyber Laws, Universal Law Publishing Co., New Delhi
4. Farouq Ahmed, Cyber Law in India, New Era publications, New Delhi

CSO XXXX**APPLIED CRYPTOGRAPHY****[3 0 3 4]****Abstract Syllabus:**

ASYMMETRIC ENCRYPTION: Asymmetric key generation techniques – Public key cryptographic systems, Applications of asymmetric encryption methods – RSA and variants, Attacks on RSA, Rabin Cryptosystem, Elgamal Cryptosystem, Elliptic Curve Cryptography, Homomorphic encryption and types. Key Establishment and Key Management, Diffie-Hellman Key Exchange, Attacks on Diffie Hellman, Digital Signatures, Elgamal Digital Signature Scheme, RSA Digital Signatures Scheme, Schnorr Digital Signature Scheme, NIST Digital Signature Algorithm, Elliptic Curve Digital Signature Algorithm, RSA-PSS Digital Signature Algorithm, Authentication Protocol, Hash Functions and MAC: Standard hashes (MD5, SHA-1, SHA-256/384/512, RIPEMD-160), Birthday Attack, Collision freeness and recent attacks, Message Authentication Code (MAC) Algorithms. Network and Internet Security: Transport Layer Security, IP Security, Email Security, SECURITY PRACTICE & SYSTEM SECURITY

Abstract Syllabus (Integrated lab):

Implementation and demonstration of asymmetric cryptographic systems, Demonstration of hashing and message digest techniques. Design and implementation of homomorphic encryption techniques. Demonstration and implementation of secure communication using standard crypto libraries. Implementation of smart card-based server/client applications Demonstration of authentication techniques. Developing cryptographic algorithms for industrial applications. Developing cryptographic algorithms for innovative applications.

References :

1. William Stallings, Cryptography and Network Security: Principles and Practice, (7e), Prentice Hall, 2017.
2. Behrouz A. Forouzan and Debdeep Mukhopadhyay, Cryptography and Network Security,(2e), McGraw Hill, 2008.

OTHER COURSES WITH LABS

CSS XXXX

PRINCIPLES OF CRYPTOGRAPHY

[3 0 3 4]

Abstract syllabus (theory):

Computer Security Concepts, Attacks, Services, Mechanisms, Classical Encryption Techniques - Symmetric Cipher Model, Substitution and Transposition Techniques, Block Ciphers and DES, Strength of DES, Block Cipher Design Principles. Advanced Encryption Standard, Block Cipher Modes of Operation, Pseudorandom Number Generation, Stream Ciphers, RC4, Euclidean Algorithm, Modular Arithmetic, Prime Numbers, Fermat and Euler theorems, Testing for Primality, Chinese Remainder theorem, Discrete Logarithms, Public Key Cryptography-Principles, RSA, D-H Key Exchange, ElGamal System, Elliptic Curve over primes, ECC, Cryptographic Hash Functions. Requirements, SHA3, Hash Functions based on Cipher Block Chaining, Message Authentication Codes, MACs Based on Hash Functions and Block ciphers, Digital Signatures -RSA, ECDSA, ElGamal, Schnorr, Digital Signature Scheme, RSA-PSS Digital Signature Algorithm

Abstract syllabus (Integrated lab) :

Implement number theoretic algorithms such as Euclidian, Inverses in congruences, Chinese Remainder Theorem, Modular Exponentiation, Classical ciphers, modern symmetric key ciphers asymmetric key ciphers, Hashing, MAC and digital signature algorithms. C/C++ could be used for implementation of the algorithms.

References:

1. William Stallings, *Cryptography and Network Security: Principles and Practice*, (8e), Prentice Hall, 2023.
2. Behrouz A. Forouzan and Debdeep Mukhopadhyay, *Cryptography and Network Security*, (3e), McGraw Hill, 2015.
3. Atul Kahate, *Cryptography and Network Security*, (4e), Tata McGraw-Hill Publishing, 2019.
4. Johnathon Katz and Yehuda Lindell, *Introduction to Modern Cryptography*, (3e), CRC Press, 2021.
5. Bruce Schneier, *Applied Cryptography-Protocols, Algorithms, and source code in C*, (2e), John Wiley & Sons, Inc., 2017.
6. https://onlinecourses.nptel.ac.in/noc25_cs31/preview Foundations of Cryptography.

Abstract syllabus (theory):

Evolution from an art form to an engineering discipline, Software development Projects, Exploratory style of software development, Emergence of software Engineering, Notable changes in software development practices. Computer Systems Engineering. Waterfall model and its extensions, Rapid Application Development, Agile development models, Spiral Model, A Comparison of different Life Cycle models, Cohesion and coupling, Layered arrangement of modules, Approaches to software design. Overview of SA/SD methodology, Structured analysis, Developing the DFD Model of a system, Structured Design,. Basic object-orientation concepts, UML diagrams, Use case model, Class diagrams, Interaction diagrams, Activity Diagram, State chart diagram, , An Object-Oriented Analysis and Design (OOAD) Methodology, Software Documentation, Testing, Unit Testing, Black-Box testing, White-Box Testing, Debugging, Program Analysis tools, Regression testing.

Abstract Syllabus (Integrated lab):

Rational Rose, Star UML, SRS, SRS of Case Study, Code Generation, Project Implementation, Use Cases, SOLID Principles, Class Diagrams, Interaction Diagrams, Package Diagrams, Component Diagrams, Deployment Diagrams, NUnit, Junit..

References :

1. Rajib Mall, *Fundamentals of Software Engineering* (5e), PHI Learning, 2019.
2. Hans Van Vliet, *Software Engineering: Principles and Practice* (3e), Wiley India, 2012.
3. Roger S. Pressman, *Software Engineering - A Practitioner's Approach* (7e), McGrawHill International Edition, 2010.
4. Bernd Bruegge, Allen H. Dutoit, *Object-Oriented Software Engineering using UML Patterns and Java* (2e) , Pearson Publication, 2011.
5. Ian Sommerville, *Software Engineering* (9e), Addison-Wesley, 2011.
6. https://onlinecourses.nptel.ac.in/noc20_cs68/preview

Abstract syllabus (theory):

Introduction to cloud computing, evolution, characteristics, comparison with traditional computing, benefits and challenges, business impact, security considerations, deployment models: public, private, community, hybrid clouds. Service Models: IaaS, PaaS, SaaS. Virtualization Fundamentals, Hypervisors, SOA Principles, Cloud Security Model: Confidentiality, integrity, availability (CIA), Threats & Vulnerabilities, Legal & Compliance Issues: Data governance, privacy laws, risk assessment frameworks, Performance Monitoring

& Disaster Recovery, Introduction to DevOps, Benefits, Agile, DevOps Lifecycle, Principles and Practices, CI/CD, Microservices Architecture, Infrastructure as Code (IaC), Tools, Introduction to Source Version Control & Workflows, Containerizing applications with Docker, Container Orchestration using Kubernetes, Monitoring and Logging, Real-World Applications of DevOps.

Abstract Syllabus (Integrated lab);

Setting up Git and GitHub, Initializing repositories, creating branches, making commits, Managing pull requests, Building Docker images and running containers, Managing multi-container applications using Docker Compose, Deploying a containerized application on Kubernetes, Setting up GitHub actions, Configuring a GitHub pipeline for Continuous Integration, Automating code builds, Setting up monitoring for a deployed application, Integrating Prometheus for metrics collection, Visualizing metrics with Grafana

References:

1. Thomas Erl, Zaigham Mahmood, and Ricardo Puttini, *Cloud Computing: Concepts, Technology, Security and Architecture*, Pearson, 2023
2. Rajkumar Buyya, James Broberg & Andrej Goscinski, *Cloud Computing, Principles & Paradigms*, Wiley Publication, 2011
3. Mikael Krief, *Learning DevOps(2e)*, Packt Publishing, 2022
4. Karl Matthias, Sean P. Kane, *Docker: Up & Running: Shipping Reliable Containers in Production(3e)*, O'Reilly Media, 2023
5. Kelsey Hightower, Brendan Burns, Joe Beda, *Kubernetes – Up and Running*, Shroff Publishers & Distributors Pvt. Ltd, 2022
6. NIST Cloud Computing Standards
<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.500-291r2.pdf>
7. Cloud computing By Prof. Soumya Kanti Ghosh | IIT Kharagpur
https://onlinecourses.nptel.ac.in/noc21_cs14

MINOR SPECIALIZATIONS

I. MULTIMODAL DATA ANALYTICS

CSP XXXX

INFORMATION RETRIEVAL

[3 0 0 3]

Abstract Syllabus:

Boolean Retrieval Model, Term vocabulary and Postings lists, Dictionaries and tolerant retrieval, Index Construction, Blocked sort based indexing, Single Pass in memory indexing, Distributed indexing, Dynamic Indexing, Index Compression, Dictionary compression, postings compression, Vector Space Model, Parameter and zone indexes, Term frequency & weighting- Inverse document frequency, tf-idf weighting, Vector space model for scoring, Evaluation in IR, Result snippets, Relevance Feedback and Query Expansion, Latent Semantic Indexing, Web Search Basics, Near duplicates and shingling, Web Crawling and Indexes, Distributing indexes, Link Analysis, Page Rank.

References :

1. Christopher D. Manning, Prabhakar Raghavan, Hinrich Schütze, *Introduction to Information Retrieval*, Cambridge University Press, 2008.
2. Stefan Buettcher, Charles L. A. Clarke, Gordon V. Cormack, *Information Retrieval: Implementing and Evaluating Search Engines*, MIT Press, 2016.
3. David A. Grossman, Ophir Frieder, *Information Retrieval: Algorithms and Heuristics*, Springer, 2004.

CSP XXXX

GRAPH ANALYTICS

[3 0 0 3]

Abstract syllabus:

Introduction to Knowledge graphs: Definition and significance in data analysis, Key concepts: Nodes, edges, triples, and graph-based data models, Knowledge graphs vs. traditional databases, Real-world applications, **Semantic foundation:** Ontologies, taxonomies, and vocabularies, Overview of semantic web standards for graph based analytics, **Modeling and Design:** Modeling knowledge graphs, Ontology Design and Semantic Standards, **Knowledge graph construction:** Domain discovery, Named entity recognition, Web information extraction, Relation extraction, Non traditional information extraction, **Knowledge graph completion:** Instance matching, statistical relational learning, Representation learning for knowledge graph, **Accessing knowledge graphs:** Reasoning and retrieval, structured querying,

question answering, **Knowledge graph ecosystem:** linked data, Knowledge graphs and ontologies in science, knowledge graph for domain specific social impact.

References:

- 1) Mayank Kejriwal, Craig A. Knoblock, Pedro Szekely, *Knowledge Graphs: Fundamentals, Techniques, and Applications*, MIT Press, 30 Mar 2021
- 2) Grigoris Antoniou and Frank van Harmelen, *A Semantic Web Primer*, The MIT Press, 2nd Edition, 2008.
- 3) Pascal Hitzler et al, *Foundations of Semantic Web Technologies*, Chapman & Hall, 2009.
- 4) Ian Robinson, Jim Webber & Emil Eifrem, *Graph Databases*, published by O'Reilly, 2015
- 5) Jesus Barrasa, Amy E. Hodler, Jim Webber, *Knowledge Graphs*, O'Reilly Media, Inc., 2021
- 6) Jesus Barrasa, Jim Webber, *Building Knowledge Graphs*, O'Reilly Media, Inc. 2023

CSP XXXX

IMAGE AND VIDEO ANALYSIS

[3 0 0 3]

Abstract syllabus:

Introduction to Digital Image Processing: Image Formation, Sampling, and Quantization, Image representation, Color models, Color transformations; **Operations on Images** Basic Gray Level Transformations, Histogram Processing, Convolution, Linear and Non-Linear filter, Fourier Transforms, Morphological Image processing; **Feature extraction and Matching:** Edge and Corner detection, Scale Invariant Features, Feature Descriptors, Feature Matching; **Image Segmentation:** Thresholding, Region-based Segmentation, Color-Based and Texture-Based Image Segmentation, Advanced Image Segmentation Techniques; **Introduction to Digital Video Processing:** Video representation and formats, Compression Techniques, Motion Estimation and Optical flow, Background subtraction; **Video Segmentation and Tracking:** Background Modeling, Motion Segmentation; Tracking algorithms; Kalman Filter, Mean-shift; **Video Analysis and Action recognition:** Scene understanding, motion-based recognition, and video-based rendering. **Case studies on:** Medical Image Segmentation, Autonomous Driving, Traffic Monitoring, Human Activity Recognition.

References:

1. Rafael C. Gonzalez, Richard E. Woods, *Digital Image Processing*, (4e), Pearson Education, 2018.
2. Richard Szeliski, *Computer Vision: Algorithms and Applications*, (2e), Springer, 2022.
3. David A. Forsyth, Jean Ponce, *Computer Vision: A Modern Approach*, (2e), Prentice Hall, 2002.

4. A Murat Tekalp, *Digital Video Processing*, Prentice Hall, 2nd Edition, 2015.
5. Mark Nixon and Alberto S. Aquado, *Feature Extraction & Image Processing for Computer Vision*, Third Edition, Academic Press, 2012
6. https://onlinecourses.nptel.ac.in/noc25_ee13, Computer Vision and Image Processing - Fundamentals And Applications by Prof. M. K. Bhuyan, IIT Guwahati

CSP XXXX

SOCIAL NETWORK ANALYSIS [3 0 0 3]

Abstract syllabus:

Introduction to Social Web, Nodes, Edges, Basic definitions, Types of networks, Layouts, Visualizing network features, Network structures and Measures, The role of Tie strength, Measuring Tie strength and its network structures, Link prediction, entity resolution, Propagation in networks, Introduction to social influence, Influence related statistics, social similarity and influence, Homophily, Existential Test for social influence, influence maximization in viral marketing, Introduction to community discovery, Types of communities, Community detection methods, The Kernighan-Lin algorithm, Agglomerative algorithms, spectral algorithms, multi-level graph partitioning, Evaluation of community detection methods, Anomaly detection in networks, Large-scale social network analysis applications and case studies, Emerging trends and issues.

References :

1. Jennifer Golbeck., *Analysing the Social Web*, Morgan Kaufmann publications, 2013
2. Charu C. Aggarwal, *Social Network Data Analytics*, Springer publications, 2011
3. Chakraborty T, *Social Network Analysis*, Wiley India Pvt. Ltd., 2021
4. David Easley and Jon Kleinberg ,*Networks, Crowds and Markets*, Cambridge University Press, 2010
5. John Scott, *Social Network Analysis*, (3e), Sage publications limited, 2013
6. <https://nptel.ac.in/courses/106106239> , **NOC:Social Network Analysis**
7. <https://nptel.ac.in/courses/106106169>, **NOC:Social Networks**

II. HEALTH CARE ANALYTICS

CSP XXXX

HEALTH INFORMATICS

[3 0 0 3]

Abstract syllabus:

Fundamentals to Healthcare Informatics: Data, Information, and Knowledge Hierarchy; Types and Properties of Healthcare Data; Data Quality and Cleaning; Healthcare Databases and its implementation; Data Warehousing in Healthcare; **Health care information system:** Meta-Modelling - Three-Layer Graph-Based Metamodel (3LGM2) – Domain Layer, Logical Tool

Layer and Physical Tool Layer; Electronic Health Records; Application Systems - Patient Administration Systems, Medical Documentation and Management Systems, Computerized Provider Order Entry (CPOE) Systems, Radiology Information Systems (RIS) and PACS; Architectural Styles in Healthcare Informatics; Aspects of Interoperability – Health Level 7; Standards - Clinical Document Architecture (CDA), DICOM, Clinical Data Interchange Standard Consortium; Integrating Healthcare Enterprise; **Security and Privacy of Healthcare Informatics:** Ethical consideration in Healthcare; Privacy law -HIPPA; Protected Health Informatics; Security Measures: De-identification of Healthcare Data, Data Compression Technique and its applications; **Application of Healthcare Informatics:** Introduction to Telemedicine; Disease Management – Role of informatics in managing chronic diseases; Mobile Health (mHealth) – Tools, trends, and future directions.

References:

1. Hoyt, R. E., & Yoshihashi, A. K. (2014). Health Informatics: practical guide for healthcare and information technology professionals. Lulu. com.
2. Wager, K. A., Lee, F.W., & Glaser, J. P. (2021). Health care information systems: a practical approach for healthcare management. John Wiley & Sons.
3. McCaffery, P. (2020). An introduction to healthcare informatics: building data-driven tools. Academic Press.
4. Nelson, R., & Staggers, N. (2013). Health Informatics: An interprofessional approach. Elsevier Health Sciences.
5. <https://nptel.ac.in/courses/110104095> Economics of Health and Health Care, IIT Kanpur
6. <https://www.medvarsity.com/courses/certificate-course-in-healthcare-informatics> Healthcare Informatics

CSP XXXX

BIOINFORMATICS

[3 0 0 3]

Abstract syllabus:

Introduction to Bioinformatics, Central dogma of biology, Digital code of life, Database sequence search: BLAST and FASTA, The modular nature of proteins, Optimal alignment methods-Simple alignments, Global and Local Alignments, Gap Penalties, Substitution patterns, Molecular Clocks, Phylogenetics, Distance Based methods for Phylogenetics, Character based methods for Phylogenetics, Parsimony, Molecular Phylogenies, Genomics and Gene Recognition: Prokaryotic Genome, Eukaryotic Genomes. Amino acids, algorithms for Protein folding, Protein and RNA Structure Prediction, Artificial Intelligence and Machine Learning in Bioinformatics, Drug Discovery.

References:

1. Dan E. Karne, Michael L Raymer, Fundamental Concepts of Bioinformatics, 2006, Pearson Education, Springer, 2024
2. Vijai Singh Ajay Kumar, Advances in Bioinformatics, Springer Nature, 2024
3. Arthur M. Lesk. Introduction to Bioinformatics, Oxford University Press, 2002
4. Stuart M. Brown. BIOINFORMATICS: A biologists guide to biocomputing and the internet, NYU Medical Center, 2000
5. <https://nptel.ac.in/courses/102106065> Bioinformatics: Algorithms and Applications

CSP XXXX

MEDICAL IMAGE PROCESSING

[3 0 0 3]

Abstract syllabus:

Digital Image Acquisition: X-Ray Imaging, Magnetic Resonance Imaging, Ultrasound. Image Enhancement: Measure of Image Quality, Contrast, Resolution and Edge enhancement, Noise Reduction. Feature Detection: Edge Tracking, Hough Transform, Corners, Blobs, SIFT and SURF Features, MSER, Saliency and Gist, Bag of Features. Segmentation: Principles and Basic Techniques, Segmentation in Feature Space, Segmentation as a Graph Problem. Registration and Normalization, Classification and Clustering, Validation: Measure of Quality, The Ground Truth. Applications of Deep learning to medical image analysis. Case studies: Disease classification, lesion and organ segmentation.

References:

1. Introduction to Medical Image Analysis, Rasmus R. Paulsen , Thomas B. Moeslund
2. Guide to Medical Image Analysis Methods and Algorithms, Klaus D. Toennies
3. David A forsyth & Jean ponce Computer vision – A modern Approach, Prentice Hall, Pearson Education India; Edition: Second
4. R. C. Gonzalez, R. E. Woods. Digital Image Processing. Pearson, Inc., Edition-Fourth
5. Bernd Jahne and Horst HauBecker, Computer vision and Applications, Academic press, 2000.
6. https://onlinecourses.nptel.ac.in/noc22_bt34/preview **Medical Image Analysis**

CSP XXXX

BIOSTATISTICS

[3 0 0 3]

Abstract syllabus :

Foundations of Biostatistics: Data and types of data, Scale of measurements, Introduction to data distributions. Population, Sample, Central Limit Theorem, Standard Error, Confidence

Intervals. Measures of central tendency, Standard normal curve, Measure of dispersion, Review of descriptive statistics, Summarizing Data, Data Presentation techniques. Hypothesis framing, Role of probability in decision making, test of significance, p-values and statistical inference. Inferential statistics, Parametric and Non-parametric, Correlation tests. Epidemiological Study Designs: observational, Experimental study designs, Clinical trials, Screening designs: Fractional factorial designs, Plackett-Burmann screening designs. Sampling and Sample size: Probabilistic sampling techniques, Non- Probabilistic Sampling techniques, Estimation of sample size: Descriptive studies and Comparative studies. Applied Regression Analysis and Survival Analysis: Multiple regression analysis, logistic regression analysis, Cox Proportional Hazards Regression Analysis, Survival analysis- Censoring, survival function, survival curve.

References:

1. Sullivan, L.M., Essentials of biostatistics for the health sciences, 3rd edition, Jones & Bartlett Learning, 2018.
2. Machin, Campbell and Walters, Medical Statistics, 4th edition, Wiley, 2007.
3. Motulsky, H., Intuitive Biostatistics: A nonmathematical guide to statistical thinking, 3rd edition, Oxford University Press, New York, 2014.
4. Utts, J and Heckard, R., Mind on statistics, 5th edition, Cengage Learning, USA.
5. David S. Moore , William I. Notz, Michael A. Fligner , Statistics in practice , W. H. Freeman publication, 1st edition, 2014.
6. Antonisamy B, Premkumar PS, Christopher S. Principles and Practice of Biostatistics- E-book: Principles and Practice of Biostatistics-E-book. Elsevier Health Sciences; 2017 May 8.
7. https://onlinecourses.nptel.ac.in/noc20_bt11/preview Biostatistics and Design of experiments By Prof. Mukesh Doble | IIT Madras.

PROGRAM ELECTIVES

CSP XXXX

IoT AND ITS APPLICATIONS

[3 0 0 3]

Abstract Syllabus :

Introduction to IoT – Definition & Evolution, Physical and Logical Design, Functional Blocks, Machine-to-Machine (M2M) Communication, and Layered Architecture, Protocols and Standards – IEEE 802.15.4, LoRaWAN, ZigBee, NB-IoT, IPv6, 6LoWPAN, Constrained Networks, MQTT, CoAP, HTTP, AMQP, DDS, and OPC-UA, Hardware Platforms and Data Acquisition Techniques – Raspberry Pi, Arduino, ESP8266, Various Sensors, Real-Time data Processing and Management, Cloud Infrastructure for IoT, Edge Computing, Fog computing and Everything-as-a-Service (XaaS) Models, IoT Networking and Power Efficiency – LPWAN and BLE, IoT Applications – Smart Homes, Industrial Automation, Industry 4.0, Smart Agriculture, Healthcare, Smart Cities, Security, Wearable Devices, and Urban Development.

References:

1. "Internet of Things: Principles and Paradigms" – Rajkumar Buyya, Amir Vahid Dastjerdi (Morgan Kaufmann, 2016)
2. "IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things" – David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton, Jerome Henry (Cisco Press, 2017)
3. "The Internet of Things: Key Applications and Protocols" – Olivier Hersent, David Boswarthick, Omar Elloumi (Wiley, 2012)
4. "Internet of Things: A Hands-On Approach" – Arshdeep Bahga, Vijay Madisetti (Universities Press, 2015)
5. "Internet of Things: Architecture, Design Principles and Applications" – Rajkamal (McGraw Hill Higher Education)
6. "Cloud Computing and IoT: A Technology Perspective" – Velte, Toby, Anthony Velte (McGraw Hill, 2019)
7. "Industrial IoT (IIoT): Concepts, Methodologies, Tools, and Applications" – IGI Global
8. "Smart Cities, Smart Future: Showcasing Tomorrow" – Mike Barlow, Cornelia Levy-Bencheson
9. "IoT in Healthcare and Ambient Assisted Living" – Zaigham Mahmood
10. "LPWAN Technologies for IoT and M2M Applications" – Benoît Hilt, Cuong Pham (Elsevier, 2020)
11. https://onlinecourses.nptel.ac.in/noc22_cs53/preview (Nptel Course)

CSP XXXX

AUGMENTED AND VIRTUAL REALITY

[3 0 0 3]

Abstract Syllabus:

Introduction of Virtual and Augmented reality: Definition, scope History of Early VR and Commercial VR Technologies, VR Industry Growth and Applications, The 3 I's of Virtual Reality (Immersion, Interaction, Imagination), Differences Between AR, VR, MR, and XR, AR/VR System Components and Interaction: Components of a VR System: Hardware & Software, AR Display Technologies: Optical See-Through & Video See-Through Displays, Multimodal Interfaces: Gesture, Haptic, and Auditory Interfaces, Perception Requirements and Characteristics in AR/VR, Input Devices: 3D Position Trackers, Navigation & Manipulation Interfaces, Output Devices: Graphics Displays, Sound Displays, Haptic Displays, Computer Vision and Tracking for Augmented Reality: Natural Feature Tracking: Marker-Based and Markerless AR, Simultaneous Localization and Mapping (SLAM) in AR, Optical and Sensor-Based Tracking, Outdoor Tracking Techniques in AR, Real-Time Rendering Pipeline for AR Applications, VR System Architectures and Modeling: Computing Architectures for VR: Workstation-Based and Distributed Systems, Parallel Processing for Real-Time VR Applications, Graphics pipeline - Overview and Transformations, Geometric and Kinematics Modeling for VR Environments, Physical and Behavioral Modeling in VR, The Rendering

Pipeline in VR and Optimization Techniques, AR/VR Development and Applications: AR/VR Development Tools: Unity3D, Unreal Engine, Vuforia, ARKit, ARCore, Application Development Process: Scene Creation, Interactions, and Physics, Programming for AR/VR: Scripting in C# (Unity) and Blueprints (Unreal), Use Cases: Gaming, Healthcare, Education, Engineering, Retail, and Entertainment, Ethics and Challenges in AR/VR Development, Future Trends and Research in AR/VR: Cloud-Based and 5G-Enabled AR/VR Applications, AI and Machine Learning in AR/VR Experiences, AR/VR in the Metaverse and Social Media Integration, Haptic Feedback and Brain-Computer Interfaces (BCI), Research Challenges and Opportunities in AR/VR

References:

1. Dieter Schmalstieg & Tobias Hollerer, *Augmented Reality: Principles & Practice*, 2nd Edition, Addison-Wesley, 2023.
2. Steven M. LaValle, *Virtual Reality*, Cambridge University Press, 2021.
3. Jesse Glover & Jonathan Linowes, *Complete Virtual Reality and Augmented Reality Development with Unity*, Packt Publishing, 2021.
4. Grigore C. Burdea & Philippe Coiffet, *Virtual Reality Technology*, 2nd Edition, Wiley-IEEE Press, 2006.
5. Tony Parisi, *Learning Virtual Reality: Developing Immersive Experiences and Applications*, O'Reilly Media, 2015.
6. Steve Aukstakalnis, *Practical Augmented Reality: A Guide to Technologies and Human Factors*, Addison-Wesley, 2016.
7. **NPTEL Online Course: Virtual Reality and Augmented Reality** – <https://nptel.ac.in/courses/106106138>

CSP XXXX

CONVERSATIONAL AI (NLP)

[3 0 0 3]

Abstract syllabus (theory):

Introduction to Conversational AI & NLP: Overview of Conversational AI and its applications, Basics of Natural Language Processing (NLP), History and evolution of chatbots and virtual assistants, Components of a Conversational AI system, Rule-based vs. ML-based vs. Hybrid chatbots. Conversational AI Architectures: Sequence-to-sequence models and their limitations, Attention Mechanism in Conversational AI, Transformer-based architectures for dialogue (BERT, GPT, T5), Dialogue Management Systems. Advanced NLP Techniques for Chatbots: Transfer Learning in NLP, Intent Recognition & Slot Filling, Dialogue Management Systems (Rule-based vs. ML-based), Enhancing Conversational AI: Speech-to-Text & Text-to-Speech Integration, Context Handling and Multi-Turn Dialogues, Custom NLU pipelines and integrating with APIs. Evaluation, Challenges, and Ethics: Evaluation Metrics for

Conversational AI, Challenges in Conversational AI and Future Trends. End-to-end conversational AI Development, Deployment.

References:

1. Daniel Jurafsky & James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, (3e), Pearson, 2024.
2. Andrew R. Freed, Conversational AI: Chatbots that work, Manning, 2021
3. Steven Bird, Ewan Klein and Edward Loper, Natural Language Processing with Python, (1e), O'Reilly Media, 2009
4. Steven Bird, Ewan Klein, Edward Loper, Natural Language Processing with Python – Analysing Text with natural language toolkit, O'Reilly Media, 2009
5. Chris Manning, Hinrich Schutze, Foundations of Statistical Natural Language Processing, MIT Press, Cambridge, 1999.
6. NPTEL: Applied Natural Language Processing By Prof. Ramaseshan R | CMI

CSP XXXX

FUNDAMENTALS OF iOS

[3 0 0 3]

Abstract syllabus :

Swift Lessons: Installations, iCloud Service, Introduction to Swift and Playgrounds, Constants Variables and Data Types, Operators and Control Flow, Strings, Functions, Structures, Structures, Classes and Inheritance, Collections, Loops, Optionals, Type Casting and Inspection, Guard, Scope, Enumerations, Constants.

SDK Lessons: Documentation, Building, Running and Debugging an app, Interface Builder Basics, Strings, Introduction to UIKit, Displaying Data, Controls in Action, Auto Layout and Stack Views, Segues and Navigation Controllers, Tab Bar Controllers, View Controller Life Cycle, Building Simple Workflows

Guided Projects: Light, about me, Calculator, Apple pie, personality quiz.

References: Develop in Swift Fundamentals, Xcode 13, by Apple Corporation, USA

CSP XXXX

iOS APPLICATION DEVELOPMENT

[3 0 0 3]

Abstract syllabus :

Tables and Persistence: Protocols, App life Cycle, MVC, Scroll views, table views, intermediate table views, system view controllers, saving data, complex input screens.

Working with Web: Closures, Extensions, Practical Animations, working with HTTP, URL session, JSON and Concurrency

Advanced Data display: Swift generics, collection views, dynamic data, compositional layout, advanced layout

Guided projects: List, Restaurant, Habits.

References:

1. Develop in Swift Data Collections, Xcode 13, by Apple Corporation, USA.

CSP XXXX

BLOCKCHAIN TECHNOLOGY

[3 0 0 3]

Abstract syllabus:

Introduction to Blockchain: Definition, History and evolution of Blockchain, Features of Blockchain, Blockchain Architecture, Types of Blockchain, Potential of Blockchain and Myths, Decentralization in Practice and Decentralized Web. Cryptographic Primitives, Blockchain Data Structure and Consensus Mechanisms. Overview of Ethereum and its architecture, Ethereum Virtual Machine, Smart contracts and its development environment, Smart Contracts with Remix IDE, Overview of DApps, DApps development. Hyperledger and Permissioned Blockchain: Overview of Hyperledger frameworks, tools and building blocks, Hyperledger Fabric Model, Chaincode. Case Studies: Application of Blockchain in finance, healthcare, agriculture, IoT, supply chain, logistics, smart grid, smart cities, data management and e-governance.

References:

1. Kumar M V, Manoj, Annappa B, Thomas Likewin, Addya Sourav Kanti, and Niranjnamurthy M. *Blockchain Technology and Applications*(1e), CRC Press, 2024.
2. S. Shukla, M. Brahmabhatt, P. Tiwari, *Blockchain Technology and Applications*, Wiley, 2020.
3. Bashir, Imran, *Mastering Blockchain: A Deep Dive into Distributed Ledgers, Consensus Protocols, Smart Contracts, DApps, Cryptocurrencies, Ethereum, and More*(3e), Packt Publishing, 2020.
4. Ritesh Modi, *Solidity Programming Essentials: A Beginner's Guide to Build Smart Contracts for Ethereum and Blockchain*, Packt Publishing, 2018.
5. S.Kulkarni, *Hands-On Blockchain with Hyperledger: Building Enterprise-Ready Applications*, Packt Publishing, 2020.
6. https://onlinecourses.nptel.ac.in/noc22_cs44/preview Blockchain Technology & Applications.

Abstract syllabus:

Introduction to Wireless Networks: Evolution, Challenges, Introduction to Wireless Communication, Electromagnetic spectrum, Spectrum regulation, Wireless propagation, Modulation techniques, Multiple access for wireless systems, Cellular concept, Wireless services Ubiquitous Connectivity, Types of Wireless Networks, Analog Cellular Systems, AMPS, DAMPS, GSM, 3G Spectrum allocation, CDMA, WCDMA, 4G and Beyond, OFDM, Fixed Wireless Networks. IEEE 802.16, WLAN, applications, topology, requirements, Physical and MAC layers, IEEE 802.11a,b and g, Ad Hoc Networks: topology, Ad Hoc routing, VANETs, The 5G Internet, 5G Mobile Networks, SDN: IMS architecture, IMS Call Flow, services within IMS

References:

1. R Nicopolitidis et al, *Wireless networks*, Wiley, first, 2011
2. Jonathan Rodriguez, *Fundamentals of 5G Mobile Networks*, Wiley, 2015
3. Kaveh Pahlavan, Prashant Krishnamurthy, "*Principles of Wireless Networks*", Prentice Hall, 2011
4. William Stallings, "*Wireless Communication and Networking*", PHI, 2014
5. <https://archive.nptel.ac.in/courses/106/105/106105160/> Wireless Networks

Abstract syllabus:

Introduction to Quantum computation, Quantum bits, Single qubit operations, Postulates of quantum mechanics, Quantum Measurement, Bell states, EPR Paradox, No Cloning Theorem, Quantum Gates, Single qubit gates, Pauli Gates, Hadamard gate, Quantum Circuits, Multi-qubit gates, CNOT gate, Toffoli Gate, Fredkin Gate, Universal quantum gates, Quantum Key Distribution, Superdense coding and Quantum Teleportation, Quantum Parallelism and entanglement, The quantum Fourier transform (QFT), Walsh-Hadamard transformation, Quantum search algorithms, Grover's Search Algorithm, Deutsch Algorithm, Deutsch-Jozsa Algorithm, Bernstein-Vazirani Algorithm, Simon's Algorithm, Shor's Factorization algorithm, Quantum error correcting codes. Overview of Qiskit- IBM quantum computing open-source tool, SDN: Designing quantum circuits and implementing quantum algorithms using Qiskit.

References :

1. M. Nakahara and T Ohmi, "Quantum Computing From Linear algebra to Physical Realizations" CRC press 2008.

2. Michael A Nielsen, and Isaac L. Chuang “Quantum Computation & Quantum Information”, (10e), Cambridge University Press, 2011.
3. https://onlinecourses.nptel.ac.in/noc19_cy31/preview Quantum Computing

CSP XXXX

REINFORCEMENT LEARNING

[3 0 0 3]

Abstract syllabus (theory):

The Reinforcement Learning Problem: Reinforcement Learning, Elements, Limitations and Scope, Multi-arm Bandits, Finite Markov Decision Processes: The–Environment Interface, Goals and Rewards Returns, Unified Notation for Episodic and Continuing Tasks, The Markov Property, Decision Processes, Optimality, Dynamic Programming - Policy Evaluation, Improvement, Iteration, Asynchronous Dynamic Programming, Generalized Policy Iteration, Efficiency, Monte Carlo Methods and Temporal-Difference Learning: Monte Carlo Prediction, Estimation of Action Values, Control, Off-policy Prediction via Importance Sampling, Incremental Implementation, TD Prediction, Advantages, Optimality, TD Control, Q-Learning, Approximate Solution Methods On-policy Approximation, Value Prediction, Gradient-Descent Methods, Linear Methods, Introduction to Deep Reinforcement Learning, Actor-Critic Methods, Applications

References:

1. Sutton, R. S., & Barto, A. G., Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, 2020
2. Stefano V. Albrecht, Filippos Christianos, Lukas Schfer, Multi-Agent Reinforcement Learning: Foundations And Modern Approaches, MIT Press, 2024
3. Warren B. Powell, Reinforcement Learning and Stochastic Optimization: A Unified Framework for Sequential Decisions, Wiley & Sons, 2022
4. Maxim Lapan, Deep Reinforcement Learning Hands-On, Packt, 2018
NPTEL: Reinforcement Learning, By Prof. Balaraman Ravindran | IIT Madras

CSE XXXX

MOBILE APPLICATION DEVELOPMENT

[3 0 0 3]

Abstract Syllabus:

Understanding Flutter and its features, Introduction to Dart programming language, Widgets and their properties, Building layouts with Flutter, Handling user input and gestures, Navigation stacks and routes, Passing data between screens, Implementing bottom navigation bars and drawer menus, Understanding state in Flutter, Managing state with setState, Using state management libraries like Bloc, Making HTTP requests in Flutter, Working with RESTful APIs, Storing data locally with shared preferences, Using SQLite for local database storage,

Implementing data caching strategies, Introduction to Firebase services, Authentication with Firebase Auth, Working with Firestore for real-time database, Integrating Firebase Cloud Messaging for push notifications, Writing unit tests and widget tests, Debugging techniques and tools in Flutter, Performance profiling and optimization strategies, App distribution.

References:

1. R. Payne, *Flutter App Development: How to Write for iOS and Android at Once*, 2nd ed. Razeware LLC, 2023.
2. raywenderlich Tutorial Team, M. Katz, K. D. Moore, and V. Ngo, *Flutter Apprentice (Third Edition): Learn to Build Cross-Platform Apps*. Razeware LLC, 2022.
3. T. Bailey and A. Biessek, *Flutter for Beginners - Third Edition: Cross-Platform Mobile Development from Hello, World! to App Release with Flutter 3.10+ and Dart 3.x*, 3rd ed. Packt Publishing, 2023.
4. M. L. Napoli, *Beginning Flutter: A Hands-On Guide to App Development*. Wiley, 2019
5. *Flutter Documentation*. [Online]. Available: <https://docs.flutter.dev>.

CSP XXXX FOUNDATIONS AND ETHICS OF GENERATIVE AI [3 0 0 3]

Abstract syllabus:

Introduction; Generative Modeling, Deep Learning, Variational Encoders, Generative Adversarial Networks, Autoregressive Models, Normalizing Flow Models, Energy-Based Models, Diffusion Models, Transformers, Music Generation, World Models, Multimodal Models, Ethical Issues in Generative AI. Ethical vs unethical use of Generative AI, professional and creative integrity while using AI to produce work. Complying with copyright laws, intellectual property. Effect of Data privacy on content generation. Transparency. Eliminating bias in outputs. Third-party fact-checking. Tangible benefits and risks of ethical GenAI use.

References:

1. David Foster, *Generative Deep Learning: Teaching Machines to Paint, Write, Compose and Play*, 2nd Edition, O'Reilly, 2023.
2. Jakub Langr, and Vladimir Bok, *GANs in Action: Deep Learning with Generative Adversarial Networks*, Manning, 2019
3. Joseph Babcock, and Raghav Bali, *Generative AI with Python and TensorFlow 2: Create images, text, and music with VAEs, GANs, LSTMs, Transformer models*, Packt, 2021.
4. Altaf Rehmani, *Generative AI for Everyone*, Bluerose Publishers Pvt. Ltd, 2024

5. Oliver Caelen and Marie-Alice Blete, *Developing Apps with GPT-4 and ChatGPT*, O'Reilly, 2023.
6. Generative AI and Large Language Models (8 weeks) By Naveen Kumar Bhansali | Indian Institute of Management Bangalore (IIMB).
https://onlinecourses.swayam2.ac.in/imb24_mg116/preview
7. Introduction to Large Language Models (LLMs) (12 weeks) By Prof. Tanmoy Chakraborty, Prof. Soumen Chakraborti | IIT Delhi, IIT Bombay.
https://onlinecourses.nptel.ac.in/noc25_cs45/preview

CSP XXXX

PROMPT ENGINEERING

[3 0 0 3]

Abstract syllabus:

Module-1: Introduction to Prompt Engineering and Its History, Principles of Prompt Design, Emotion and Tone Management in Prompt Design, User-Centered Design, Creative Thinking and Problem-Solving Techniques, Cultural and Linguistic Diversity in Prompt Design, Ethics and Responsibility in Prompt Engineering, Accessibility and Inclusivity in Prompt Design, Prompt Types and Custom Prompt Techniques: Open-Ended, Close-Ended, Instructional, and Contextual Prompts, Prompt Techniques: Zero-Shot, Few-Shot, Chain-of Thought, Tree of Thoughts Prompts and Fine-Tuning LLM Responses, Persuasion and Rhetorical Techniques in Prompt Engineering, Communication Strategies in Prompt Engineering, Feedback and Revision for Prompt Analysis, Evaluation and Development Suggestions on Prompts. Module-2: Contextual Understanding, Advanced Applications of Prompt Engineering for Understanding and Crafting Specialized Prompts for Industry Specific Applications, Enhancing Learning through Effective Educational Prompts, Ensuring Compliance and Accuracy in Legal and Regulatory Prompts, Prompt Engineering Applications For Creating & Engaging Prompts for Media, Marketing, and Advertising, Understanding Psychological and Sociological Factors in Prompt Design, Algorithmic Thinking and Structural Design in Prompts, Data and Information Management in Prompt Design, Multimodal Prompts: Integrating Multiple Data Sources for Enhanced Interactions, Visual, Musical, and Auditory Prompts, Prompts for Risk and Crisis Management, Prompts for Global Communication and Collaboration, Evaluation and Testing Processes in Prompt Design, Preparation for Innovative Technologies and Trends in the Future of Prompt Design and Development.

References:

1. Zisan Cihangir ISIN, Hilal FIDAN, Tamer ISIN, Dr. Mustafa Kemal TOPCU, Prompt Engineering 101, Amazon.com, Inc, 2024
2. Nathan Hunter, The Art of Prompt Engineering with ChatGPT A Hands-On Guide (Learn AI Tools the Fun Way!), Amazon.com, Inc , 2023

3. James Phoenix and Mike Taylor, Prompt Engineering for Generative AI, O'Reilly Media Inc, 2024
4. Russel Grant, PROMPT ENGINEERING AND CHATGPT, Amazon.com, Inc, 2024
5. <https://www.udemy.com/course/mastering-ai-powered-prompt-engineering-with-ai-models/?couponCode=NVDIN35> Prompt Engineering [Master Prompt Engineering In 2025 With New AI Models]
6. <https://www.coursera.org/learn/prompt-engineering> Prompt Engineering [Prompt Engineering for ChatGPT].

CSP XXXX

PRODUCT MANAGEMENT

[3 0 0 3]

Abstract syllabus:

Introduction to Product Management, Market Research and Customer Insights, Ideation and Concept Development, User-Centered Design and Usability Testing, Product Requirement definition, Roadmapping and Prioritization, Product Development and Agile Execution, Pricing Strategies and Revenue Models, Product Metrics and Success Measurement, Taking Product to Market, Post-Launch Analysis and Iteration, Cross Functional Collaboration and Leadership, Ethics in Product Management, Product Strategy and Competitive Analysis, Product Marketing and Customer Acquisition, Product Lifecycle Management and Growth

References:

1. Karl T. Ulrich & Steven D. Eppinger, "Product Design and Development", (7e), McGraw Hill, 2020.
2. Hans-Bernd Kittlaus, Software Product Management: The ISPMA-Compliant Study Guide and Handbook, (2e), Springer, 2022
3. David A. Aaker, Strategic Market Management (12e), Wiley, 2023
4. Alistair Croll & Benjamin Yoskovitz, Lean Analytics: Use Data to Build a Better Startup Faster, (1e), O'Reilly Media, 2013.
5. Dan Olsen, The Lean Product Playbook: How to Innovate with Minimum Viable Products and Rapid Customer Feedback, (1e), Wiley 2015.
6. https://onlinecourses.swayam2.ac.in/imb19_mg01/preview New Product Development, Indian Institute of Management Bangalore

CSP XXXX

ENTERPRISE DATA ARCHITECTURE

[3 0 0 3]

Abstract syllabus:

Introduction to Enterprise Architecture:- Overview, core elements, Structure of enterprises, Introduction to Enterprise Data Architecture (EDA), Evolution of architecture, Monolithic

systems – Mainframes. N-tier Architecture:- Introduction to N-tier architecture, Application Layer, Data Layer – Structured and Unstructured Data, Communication Layer, Hands-on N-tier architecture.. Service oriented architecture and Micro services:- Service oriented architecture, Web Services, Introduction to Microservices, Components of Microservices, Containers, Orchestration, Mesh, API Design, Data Handling, Architectural principles, Effectiveness of SoA. Data Models and Data governance: - Introduction to Data Models, Performance considerations, rendering, performance testing and monitoring, Disaster Recovery strategies, Fault Tolerance and Recovery, data-sharding, de-duplication in-memory computing, Scaling, Data governance, Security, privacy, value and risk, Repository and Support Tool. Architecture for Modern Technologies: - Hardware, Polycloud, Modern communication, Architecture for AI systems Enterprise Architectural frameworks: Zachman Framework and TOGAF

References :

1. Andy Graham, *The Enterprise Data Model: A framework for enterprise data architecture*, Koios Associates Ltd, 2nd edition, 2012.
2. Charles D. Tupper, *Data Architecture: From Zen to Reality*, Morgan Kaufmann, 1 edition, 2011.
3. Scott A. Bernard , *An Introduction to Enterprise Architecture*, AuthorHouse, 3rd edition.
4. <https://www.opengroup.org/togaf>, TOGAF, The Open Group

CSP XXXX

OPTIMIZATION TECHNIQUES

[3 0 0 3]

Abstract Syllabus:

Linear Programming: Problem Formulation, Linear Programming (LP) in standard form, Graphical Solution, Simplex Method, Big M Method. Transportation and Assignment Model: Transportation problem formulation, optimal solution, unbalanced transportation problem, degeneracy, Assignment problem, Hungarian Problem. Network Analysis: Graphs, Network and Flows, Minimum cost flow Models, Sources, Sinks Max Flow - Min Cut Theorem, CPM and PERT Networks. Inventory: Introduction, Single Item, Deterministic model, Purchase Inventory model with one price, break and multiple price breaks. Dynamic Programming: Introduction, Forward and backward recursions, Bellman's Principle of Optimality. Equipment Replacement Model, Allocation Problem, Inventory Models. Decision Theory: Decision under certainty: Analytic Hierarchy Process (AHP), decision under risk: decision trees, expected value criterion, Variations of the Expected value criterion, decision under uncertainty: Laplace, MinMax, Savage, Hurwicz method. Game Theory: Introduction, Minmax – Maxmin pure strategies, Optimal solution of two person zero sum games, solution of mixed strategy games,

2 x 2 games, 2 x n games, m x 2 games. Heuristics and approximation algorithms: approximation algorithms for Travelling Salesman Problem (TSP), Vertex cover problem.

References:

1. Taha H, *Operation Research: An Introduction*, 10th Edition, McMillan, 2017.
2. Rardin, Ronald L., *Optimization in Operations Research*, Pearson Education (2005)
3. Ravindra K. Ahuja, Thomas L. Magnanti, James B. Orlin, *Network Flows: Theory, Algorithms, and Applications*, Pearson New International Edition, 2014.
4. Teofilo F. Gonzalez, *Handbook of Approximation Algorithms and Metaheuristics*, Chapman & Hall/CRC Computer and Information Science Series, 1st Edition, 2007.
5. S.S. Rao, *Engineering Optimization: Theory and Practice*, New Age International Pvt. Ltd., New Delhi, 2013.
6. NPTEL course: Optimization from fundamentals, By Prof. Ankur A. Kulkarni, https://onlinecourses.nptel.ac.in/noc21_me10/preview

CSP XXXX

CYBER FORENSICS

[3 0 0 3]

Abstract Syllabus:

Understanding the Digital Forensics Profession and Investigations: - An Overview of Digital Forensics, Preparing for Digital Investigations, Procedures for Private-Sector High-Tech Investigations. The Investigator's Office and Laboratory:- Understanding Forensics Lab Accreditation Requirements, Data Acquisition, Validating Data Acquisitions Processing Crime and Incident Scenes, Current digital forensics tools, Network Forensics and email investigation, Exploring the Role of E-mail in Investigations, Exploring the Roles of the Client and Server in E-mail, Investigating E-mail Crimes and Violations, Understanding E-mail Servers, Using Specialized E-mail Forensics Tools. Report Writing for High-Tech Investigations, Expert Testimony in Digital Investigations and ethics, Understanding the Importance of Reports, Guidelines for Writing Reports, Generating Report Findings with Forensics Software Tools, Preparing for Testimony, Testifying in Court, Preparing for a Deposition or Hearing, Preparing Forensics Evidence for Testimony, Applying Ethics and Codes to Expert Witnesses.

References:

1. Bill Nelson, Amelia Phillips, *Guide to Computer Forensics and Investigations: Processing Digital Evidence*, Fifth Edition, CENGAGE Learning, 2015.
2. John R. Vacca, *Computer Forensics, Computer Crime Scene Investigation*, 3rd Revised Edition, Jones and Bartlett Publishers, Inc., 2019.
3. Keith J. Jones, Richard Bejtlich, Curtis W. Rose, *Real Digital Forensics*, Addison Wesley Pearson Education, 2006.

4. NPTEL Course: Digital Forensics, By Dr. Jeetendra Pande,
https://onlinecourses.swayam2.ac.in/nou22_cs05/preview

CSP XXXX

SEMANTIC WEB

[3 0 0 3]

Abstract Syllabus:

Semantic Web web technologies, a layered approach, RDF-data model, syntaxes, RDFS-adding semantics, RDF schema, RDF and RDF schema in RDFS, Axiomatic schematics for RDF and RDF schema, Direct inference system for RDF and RDFS, Querying Semantic Web – SPARQL, Ontology Languages- OWL2, Ontology Design and Management, Current Applications of the Semantic Web.

References:

1. Grigoris Antoniou, Paul Groth, Frank van Harmelen, Rinke Hoekstra, and Eric Yu, A Semantic Web Primer, 3rd Edition, MIT Press, 2012.
2. Peter Szeredi, Gergely Lukacsy, Tamas Benko, and Zsolt Nagy, The Semantic Web Explained, Cambridge University Press, 2014
3. Liyang Yu, Introduction to the Semantic Web and Semantic Web Services, CRC Press, 2019
4. Elisa F.Kendall, Deborah L.McGuinness, Ying Ding, and Paul Groth, Ontology Engineering, Morgan & Claypool Publishers, 2019
5. NPTEL course: Artificial Intelligence: Knowledge Representation And Reasoning, By Prof. Deepak Khemani, https://onlinecourses.nptel.ac.in/noc24_cs14/preview

CSP XXXX

DEEP LEARNING

[3 0 0 3]

Abstract syllabus:

Introduction to Deep Learning: Motivation, History and Applications of Deep Learning, Machine Learning Vs. Deep Learning; **Fundamental of Neural Networks:** Artificial Neuron, Perceptron Model, Perceptron Learning, Multi-layer Network of Perceptrons, Artificial Neural Networks, Activation Functions, Loss functions, Gradient Descent, Backpropagation, Deep Neural Networks. **Optimization of Neural Networks:** Bias-variance trade-off, Underfitting and Overfitting, Regularization, Batch normalization, Weight initialization strategies, Learning rate schedulers, Optimizers. **Convolutional Neural Networks:** Convolution, Stride and Padding, Pooling, Properties of CNN, Classical CNN architectures for Image Classification. **Sequence Models:** RNN, LSTM, GRU, Encoder-Decoder architecture, Attention mechanism, Sequence models for Neural Machine Translation. **Representational Learning:** Stacked autoencoders, Denoising autoencoders,

Sparse autoencoders, Contractive Autoencoders, Convolutional Autoencoders, Variational Autoencoders, Deep Generative Models, Introduction to Transformers

References:

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, "*Deep Learning*", MIT Press 2016.
2. Charu C. Aggarwal, "*Neural Networks and Deep Learning*", Springer 2018.
3. Eli Stevens, Luca Antiga, and Thomas Viehmann, "*Deep Learning with PyTorch*", Manning, 2020
4. Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017
5. https://onlinecourses.nptel.ac.in/noc24_cs114, Deep Learning by Prof. Sudarshan Iyengar, IIT Ropar
6. https://onlinecourses.nptel.ac.in/noc25_ee13, Computer Vision and Image Processing - Fundamentals And Applications by Prof. M. K. Bhuyan, IIT Guwahati

CSP XXXX MACHINE LEARNING TOOLS AND TECHNOLOGY [3 0 0 3]

Abstract syllabus:

Supervised Machine Learning: Algorithms and Applications- Introduction, Supervised Machine Learning, Linear and Logistic Regression, Support Vector Machines, Decision Tree, Machine Learning Applications in Daily Life. *Zonotic Diseases Detection Using Ensemble Machine Learning Algorithms*- Introduction, Bayes Optimal Classifier, Bootstrap Aggregating (Bagging), Bayesian Model Averaging (BMA), Bayesian Classifier Combination (BCC), Bucket of Models, Stacking, Efficiency Analysis. *Model Evaluation*-Introduction, Model Evaluation, Metric Used in Regression Model, Confusion Metrics, Correlation, Natural Language Processing (NLP), Additional Metrics, Summary of Metric Derived from Confusion Metric, Metric Usage, Pro and Cons of Metrics. *The Significance of Feature Selection Techniques in Machine Learning*- Introduction, Significance of Pre-Processing, Machine Learning System, Feature Extraction Methods, Feature Selection, Merits and Demerits of Feature Selection. *Detection of Diabetic Retinopathy Using Ensemble Learning Techniques*- Introduction, Related Work, Methodology, Data Pre-Processing, Feature Extraction, Learning, Proposed Models, Experimental Results and Analysis. *Machine Learning and Deep Learning for Medical Analysis—A Case Study on Heart Disease Data*-Introduction, Related Works, Data Pre-Processing, Feature Selection, ML Classifiers Techniques, Hyperparameter Tuning, Dataset Description, Experiments and Results. *Predictive Analysis on Online Television Videos Using Machine Learning Algorithms*- Introduction, Overview of Video Analytics, Machine Learning Algorithms, Proposed Framework, Feature Selection, Classification, Online Incremental Learning, Results and Discussion. *Application of Machine Learning Algorithms With Balancing Techniques for Credit Card Fraud Detection:A Comparative Analysis*-

Introduction, Methods and Techniques, Results and Discussion. *Measuring Urban Sprawl Using Machine Learning*- Introduction, Literature Survey, Remotely Sensed Images, Feature Selection, Classification Using Machine Learning Algorithms, Results, Discussion and Conclusion.

References :

1. Pradeep Singh, *Fundamentals and Methods of Machine and Deep Learning-Algorithms, Tools and Applications*, second edition, Scrivener Publishing, WILEY, 2022
2. M. Gopal, *Applied Machine Learning*, McGraw Hill Education, 2018
3. Ethem Alpaydin, *Introduction to Machine Learning*, 2nd edition, MIT Press. 2010
4. Peter Harrington, *Machine Learning in Action*, Manning Publications, 2012.
5. Andreas C. Müller & Sarah Guido, *Introduction to Machine Learning with Python*, O'Reilly Media Inc., 2017
6. Gopinath Rebala · Ajay Ravi, Sanjay Churiwala, *An Introduction, to Machine Learning*, Springer 2019.
7. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, second edition, Springer, 2008.
8. <https://nptel.ac.in/courses/106106202> MACHINE LEARNING TOOLS AND TECHNOLOGY [Machine Learning]

CSP XXXX

WEB TECHNOLOGIES

[3 0 0 3]

Abstract Syllabus :

Introduction to World Wide Web, Protocols and Programs, Application and Development Tools, Web Design, Introduction to HTML 5, Tags and simple HTML forms, Meta tags, Frames, Style Sheets: CSS formatting, Introduction to JavaScript, Client-Side Scripting, Advanced Scripting, JavaScript Objects, JavaScript ES6, DOM and Web browser environments, DHTML, Combining HTML and CSS, Ajax, HTTP/HTTPS, Restful APIs, XML and JSON, Server-Side Scripting with PHP, NoSQL, Introduction to MongoDB, CRUD Operations.

References:

1. DT Editorial Services, *HTML 5 Black Book*, 2nd Edition, DreamTech Press, 2016
2. Powell. Thomas A., *JavaScript: The Complete Reference*
3. lemay. Laura, rafe colburn, jennifer kyrnin, *Mastering HTML, CSS & JavaScript Web*, BPB Publication, 2016
4. Vishvajeet. Sisodia, *Basic of Web Design, HTML, CSS3*, Centrum Press, 2014
5. Karl Seguin, *The Little MongoDB Book* (<https://www.openmymind.net/mongoddb.pdf>)
6. https://onlinecourses.swayam2.ac.in/nou25_cs09/preview Web Technology

Abstract syllabus :

Introduction to Game Theory, Strategic Interactions, Types of Games, Non-Cooperative Game Theory: Key Notions in Game Theory, Extensive-Form Games, Normal-Form Games, Dominant Strategy Equilibria, Pure Strategy Nash Equilibrium, Mixed Strategy Nash Equilibria, Utility Theory, Matrix Games, Bayesian Games, Cooperative Game Theory: Correlated Strategies and Correlated Equilibria, Two Person Bargaining Problem, Coalitional Games with Transferable Utility, Core of Coalition Games, Shapley Value and other Solution Concepts in Game Theory, Fundamentals of Mechanism Design, VCG Mechanisms, Introduction to Algorithmic Game Theory, Price of Anarchy, Applications in Computing.

References:

1. Y. Narahari, *Game Theory and Mechanism Design*, IISc Press and World Scientific, 2020.
2. Michael Maschler, Eilan Solan, and Schmuel Zamir, *Game Theory*, Cambridge University Press, 2nd Edition, 2020.
3. Dario Bauso, *Game Theory with Engineering Applications*, SIAM, Philadelphia, 2016.
4. Eva Tardos, Noam Nisan, Tim Roughgarden, and Vijay V. Vazirani, *Algorithmic Game Theory*, Cambridge University Press, 2007.
5. https://onlinecourses.nptel.ac.in/noc22_cs77/preview Introduction to Game Theory and Mechanism Design:

Abstract syllabus :

Introduction; Pre-model Interpretability and Explainability: EDA, Feature engineering; Model Visualization Techniques and Traditional Interpretable Algorithms: Model validation and evaluation, Classification model visualization, Traditional interpretable algorithms, Model Interpretability: Interpretable vs. explainable algorithms, Ensemble-based explainable machines, Rule-based techniques, Scoring system; Post-Hoc Interpretability and Explanations; Explainable Deep Learning: Visualizing Neural Networks, Interpretability of Transformer Models, Intrinsic, Perturbation, Gradient/Backpropagation; Fairness, Bias and Interpretability: Fairness Metrics, Bias Mitigation in AI Models; Explainability in Different Applications Domains

References:

1. Uday Kamath, and John Liu, *Explainable Artificial Intelligence: An Introduction to Interpretable Machine Learning*, Springer, 2021
2. Christoph Molnar, *Interpretable Machine Learning*, Shroff Publishers, 1st Edition, 2024
3. Wojciech Samek, Grégoire Montavon, Andrea Vedaldi, Lars Kai Hansen, and Klaus-Robert Müller, *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*, Springer, 2019
4. Serg Masis, *Interpretable Machine Learning with Python*, Packt Publishing Ltd, 2021
5. <https://www.coursera.org/specializations/explainable-artificial-intelligence-xai> (XAI)
6. https://onlinecourses.nptel.ac.in/noc24_cs132/preview (Responsible and Safe AI Systems)

CSP XXXX

USER INTERFACE ENGINEERING

[3 0 0 3]

Abstract syllabus:

Goal-Directed Design : A Design Process for Digital Products, Understanding the Problem: Design Research, Modeling Users: Personas and Goals, Setting the Vision: Scenarios and Design Requirements, Designing the Product: Framework and Refinement, Making Well-Behaved Products: A Basis for Good Product Behavior, Digital Etiquette, Optimizing for Intermediates, Orchestration and Flow, Reducing Work and Eliminating Excise, Metaphors, Idioms, and Affordances, Preventing Errors and Informing Decisions, Designing for Different Needs, Interaction Details: Designing for Mobile and Other Devices, Designing for the Web, Design Details: Controls and Dialogs

References:

1. Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel, *About Face: The Essentials of Interaction Design*, 4th Edition, Wiley, 2014.
2. Yvonne Rogers, Helen Sharp, Jennifer Preece, *Interaction Design: Beyond Human-Computer Interaction*, 6th Edition, Wiley, 2023.
3. Masaaki Kurosu (Editor), *Human-Computer Interaction. Design and User Experience Case Studies*, 1st Edition, Springer, 2021
4. Don Norman, *The Design of Everyday Things*, Revised Edition, Basic Books, 2013.
5. Steve Krug, *Don't Make Me Think: A Common Sense Approach to Web and Mobile Usability*, 3rd Edition, New Riders, 2013.

6. William Lidwell, Kritina Holden, Jill Butler, Universal Principles of Design, 2nd Edition, Rockport Publishers, 2010.
7. https://onlinecourses.nptel.ac.in/noc23_ar24/preview Introduction To Interaction Design by IIT Roorkee

CSP XXXX

BIG DATA ANALYTICS

[3 0 0 3]

Abstract Syllabus :

Overview of Big Data: Big Data Definition, Big Data Types, Analytics, Industry Examples of Big Data. Distributed and Parallel Computing for Big Data, Hadoop, Cloud Computing and Big Data, In-memory Computing Technology for Big Data, Big Data Stack, Virtualization and Big Data. Hadoop: Hadoop & Hadoop EcoSystem, Moving Data in and out of Hadoop, Inputs and outputs of MapReduce, Hadoop Architecture, HDFS, Common Hadoop Shell commands, NameNode, Secondary NameNode, and DataNode. Hadoop MapReduce paradigm, Map and Reduce tasks, Job, Task trackers , Algorithms using map reduce, Examples of Map Reduce (Word count problem, Matrix-Vector Multiplication), YARN & Zookeeper, Hadoop Cluster Setup & Hadoop Configuration. Hive Architecture, Comparison with Traditional Database, HiveQL - Querying Data - Sorting and Aggregating, Map Reduce Scripts, Joins & Subqueries, HBase concepts, Advanced Usage, Schema Design & Indexing - PIG, Zookeeper. Spark: RDD's in Spark, Data Frames & Spark SQL, Spark Streaming, MongoDB, NoSQL

References:

1. Chris Eaton, Dirk Deroos et al., “Understanding Big data”, McGraw Hill, 2012.
2. Boris lublinsky, Kevin t. Smith, Alexey Yakubovich, “Professional Hadoop Solutions”, Wiley, ISBN: 9788126551071, 2015.
3. Tom White, “HADOOP: The definitive Guide”, O Reilly 2012.
4. Aven Jeffrey, “Data Analytics with Spark Using Python”, Big Data, First Edition, Pearson Paperback, November 2018.

NPTEL:

1. Prof. Rajiv Misra, IIT Patna, “Big Data Computing”,
https://onlinecourses.nptel.ac.in/noc20_cs92/preview
2. Prof. Sandeep Singh Rawat, Indira Gandhi National Open University, New Delhi, “Data Science and Big Data”, https://onlinecourses.swayam2.ac.in/nou25_ma07/preview

Abstract Syllabus:

Introduction to graph theory, Graph types, Graph data structures, Graph storage and models, Resource Description Framework (RDF), Labelled Property Graph (LPG), Traversal Algorithms (BFS, DFS), Shortest Path Algorithms (Dijkstra, Bellman-Ford), Minimum Spanning Tree (Kruskal, Prim), Community Detection (Louvain), Graph Pattern Matching, Graph databases, Overview of Popular Graph Databases (Neo4j, TigerGraph, Amazon Neptune), Advanced Graph Technology: TigerGraph, Architecture of TigerGraph , Data Modelling in TigerGraph , Data Query Using GSQL, Data/Graph Visualizations using TigerGraph Insights, Graph Algorithms in TigerGraph, Graph Neural Networks (GNNs), Applications of Graphs in AI/ML, Graph-Based Learning Algorithms Integration of Graphs with LLMs, Graph Embedding, Industry Use Cases of TigerGraph

References:

1. Lee, Nguyen, Chang, Graph-Powered Analytics and Machine Learning with TigerGraph, O'Reilly Media, Inc, 2023.
2. Gosnell, Broecheler, The Practitioner's Guide to Graph Data: Applying Graph Thinking and Graph Technologies to Solve Complex Problems, O'Reilly Media, Inc, 2020.
3. Negro, Eifrem, Graph-Powered Machine Learning, Manning, 2021.
4. Robinson, Webber, Eifrem, Graph Databases O'Reilly Media, Inc, 2013.
5. Ma, Tang, Deep Learning on Graphs, Cambridge University Press, 2021.